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INTERNSHIP REPORT

Under the theme:

**Development of a Vehicle-To-Everything
(V2X) Communication System
for Electric Vehicles: A Renault Twizy
Case Study**

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Dedication

I primarily dedicate this humble work to my parents, who have always been an unwavering and precious source of support, love, and motivation for me to move forward. Thank you for giving me strength and believing in my dreams.

I also lovingly dedicate this work to my dear friends and teachers who have been a source of inspiration, guidance, and encouragement throughout my academic journey.

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Contents

Acknowledgements	3
List of Acronyms	11
Abstract	13
General Introduction	15
Host Organization – University UIR and LERMA Laboratory	17
1 State of the Art	23
1.1 Vehicle-to-Everything Fundamentals	23
1.1.1 Definition, Benefits, and Current Trends	23
1.2 Communication Technologies and Standards	23
1.2.1 Radio Access Technologies: DSRC vs C-V2X Evolution	23
1.2.2 Standardization and Regulatory Framework	25
1.3 Advanced V2X Application Domains	25
1.3.1 Safety-Critical and Traffic Efficiency Applications	25
1.3.2 Vehicle-to-Grid (V2G) and Energy Management	26
1.3.3 Smart City Integration and Mobility Services	26
1.4 Security, Privacy, and Trust Frameworks	26
1.4.1 Cybersecurity Challenges and Solutions	26
1.4.2 Privacy Protection and Data Management	27
1.5 Emerging Technologies and Future Directions	27
1.5.1 Artificial Intelligence and Machine Learning Integration	27
1.5.2 6G and Next-Generation Connectivity	27
1.6 Research Gaps and Implementation Challenges	28
2 System Requirements & Design	29
2.1 Project Requirements	29
2.2 Hardware Components Overview	29
2.2.1 Microcontrollers and Computing Platforms	29
2.2.2 Communication and Interface Modules	30
2.2.3 Network Infrastructure Components	31
2.3 Test Vehicle Platform	32
2.3.1 Renault Twizy Electric Vehicle	32
2.4 Communication Protocols and Technologies	35
2.4.1 ESP-NOW for V2V and V2G Communication	35
2.4.2 Wi-Fi TCP/IP for V2N Communication	37
2.5 Vehicle Data Acquisition System	38
2.5.1 Controller Area Network (CAN Bus) Fundamentals	38
2.5.2 Electronic Control Unit (ECU) Architecture	39

2.5.3	CAN Frame Structure and Data Format	39
2.5.4	Vehicle Access Through OBD-II Interface	40
2.5.5	Data Acquisition System Implementation	40
2.6	OBU and RSU System Architecture	41
2.6.1	Custom OBU Design and Implementation	41
2.6.2	OBU Physical Prototype Development	42
2.6.3	EEBLAB Custom EVCS RSU Design and Architecture	43
2.7	System Integration and Communication Flow	44
2.7.1	Summary and System Benefits	44
3	Vehicle-to-Vehicle (V2V) Implementation	45
3.1	Background on V2V	45
3.2	Proof of Concept Implementation	46
3.3	Implementation on Twizy OBU	46
3.4	Use Cases Tested & Results	51
3.5	Challenges & Limitations	53
3.6	Chapter Summary	53
4	Vehicle-To-Grid (V2G) Implementation	55
4.1	Background on V2G	55
4.2	System Architecture	56
4.3	Core Components	56
4.4	Vehicle-to-Grid Use Cases and Applications	57
4.5	Experimental Results and System Validation	61
4.5.1	Challenges and limitations	64
4.6	Chapter Summary	65
	Conclusion and Future Work	67
	Bibliography	69

List of Figures

1	Global electric car sales, 2014-2024 (IEA). The rapid growth underscores the industry shift towards electrification.	15
2	Research laboratories and centers at UIR.	18
3	LERMA Laboratory at UIR.	18
4	Key projects carried out at LERMA Laboratory.	19
5	The Energy-Efficient Building (EEB) Platform at UIR.	19
6	Key projects under the EEBLAB initiative.	20
2.1	ESP32 microcontroller development board	29
2.2	NVIDIA Jetson Nano development board	30
2.3	Raspberry Pi 3 single-board computer	30
2.4	MCP2515 CAN controller module with TJA1050 transceiver	31
2.5	PZEM-004T power measurement module	31
2.6	TP-Link M7200 4G LTE mobile hotspot	32
2.7	Renault Twizy electric vehicle	32
2.8	Renault Twizy key technical specifications	33
2.9	Detailed Twizy vehicle dimensions	34
2.10	Twizy powertrain architecture showing battery, inverter, gear motor, and wheel integration	34
2.11	Vendor-specific action frame structure utilized by ESP-NOW protocol	36
2.12	CAN bus topology and ECU communication architecture	38
2.13	Standard CAN bus termination using 120Ω resistors at network endpoints	39
2.14	Simplified ECU architecture showing MCU, CAN controller, and transceiver integration	39
2.15	Standard CAN 2.0A data frame format and field definitions	40
2.16	OBD-II connector interface for vehicle CAN bus access	40
2.17	Complete CAN data acquisition system architecture using ESP32 and MCP2515 inte- gration	41
2.18	Complete architecture of the custom OBU implemented in the Twizy vehicle	41
2.19	Physical prototype of the custom OBU system installed in the Twizy vehicle	42
2.20	EEBLAB custom Electric Vehicle Charging Station with integrated RSU	43
2.21	Detailed architecture of the EVCS RSU prototype system	44
3.1	Overview of V2V technology and its networking capabilities	46
3.2	Block diagram of the CAN data acquisition setup using ESP32 and MCP2515	46
3.3	CAN Database (DBC) creation flow	47
3.4	Open-source Twizy CAN DBC	48
3.5	Overview of CAN frames captured and their structure	48
3.6	CAN bus data parsing flow	49
3.7	CAN and GPS data captured and stored on the Jetson Nano	49
3.8	Example content of parsed and merged CAN data	50
3.9	Twizy status dashboard built on ThingsBoard	51
3.10	Twizy status update dashboard built on ThingsBoard while driving the vehicle accross UIR's compos	51

3.11	V2V communication test before implementing V2G	52
3.12	V2V communication test with V2G implemented	52
4.1	System architecture of the Twizy V2G implementation	56
4.2	EEBLAB custom Electrical-Vehicle-Charging-Slot	56
4.3	Overview of the EEBLAB custom charging station for vehicles and small devices	58
4.4	Twizy charging from EEBLAB custom EVCS's slot2	59
4.5	EVCS and Vehicle Data Transmitted to EEBLAB ThingsBoard Local Server	60
4.6	V2G Communication State Transition Diagram	61
4.7	Twizy V2G System Output Prior to Charging Session Initiation	62
4.8	EVCS System Output Prior to Charging Session Initiation	62
4.9	Twizy V2G System Output Following Charging Session Initiation	63
4.10	EVCS System Output During Active Charging Session	63
4.11	Twizy V2G System Output When Charging Session Has ended	63
4.12	EVCS System Output After Active Charging Session Has ended	64
4.13	V2G Recommendation System Architecture	64

List of Tables

1.1	Comparison of V2X Technologies: DSRC and C-V2X	24
1.2	Allocation of Spectra in Major Regions	25
2.1	12V auxiliary battery specifications	35
2.2	Traction battery technical specifications	35
2.3	Custom V2V data packet structure implemented over ESP-NOW protocol	36

List of Acronyms

- 3GPP** 3rd Generation Partnership Project – Defines LTE, 5G, and NR-V2X specifications
- 5G** Fifth Generation – Latest mobile communication standard enabling enhanced V2X capabilities
- 6G** Sixth Generation – Next-generation wireless technology expected to revolutionize V2X with ultra-low latency
- AI** Artificial Intelligence – Computer systems that can perform tasks typically requiring human intelligence
- BSM** Basic Safety Message – Standardized message format for sharing vehicle status and position data in V2X
- CAN** Controller Area Network – A robust automotive communication protocol used to connect ECUs inside a vehicle
- CAN DBC** CAN Data-Base-Container
- C-V2X** Cellular Vehicle-to-Everything – LTE/5G-based V2X communication technology (PC5 direct mode + Uu network mode)
- DSRC** Dedicated Short-Range Communications – IEEE 802.11p-based protocol for V2X
- ECU** Electronic Control Unit – A microcontroller-based unit responsible for a specific function in a vehicle
- ESP-NOW** Espressif’s proprietary low-power, peer-to-peer communication protocol for ESP32 devices
- ETSI** European Telecommunications Standards Institute – Standards body defining ITS-G5 and V2X protocols in Europe
- EV** Electric Vehicle – Vehicle powered by electric motors using energy stored in batteries
- EVCS** Electric Vehicle Charging Station – Infrastructure for charging electric vehicle batteries
- FCC** Federal Communications Commission – U.S. regulatory body that allocates spectrum for V2X
- GNSS** Global Navigation Satellite System – Satellite-based positioning system including GPS, Galileo, and GLONASS
- GPS** Global Positioning System – U.S. satellite navigation system for location and timing
- IoT** Internet of Things – Network of interconnected devices that can communicate and exchange data
- ISO** International Organization for Standardization – Global standards organization developing V2G and automotive protocols

- ITS** Intelligent Transportation Systems – Advanced applications integrating information and communication technologies
- ITS-G5** European implementation of IEEE 802.11p for intelligent transportation systems
- LTE** Long Term Evolution – 4G mobile communication standard used in C-V2X
- MEC** Multi-access Edge Computing – Computational capabilities placed at the network edge for low-latency processing
- ML** Machine Learning – Subset of AI enabling systems to learn and improve from data without explicit programming
- NR-V2X** New Radio V2X – The 5G evolution of C-V2X enabling ultra-reliable low-latency communications
- OBU** On-Board Unit – The embedded system installed in the vehicle to handle V2X communication and data processing
- OTA** Over-the-Air – Wireless delivery of software updates and configuration changes to vehicles
- PC5** Direct Communication Interface – Sidelink interface in C-V2X enabling direct vehicle-to-vehicle communication
- PKI** Public Key Infrastructure – Cryptographic framework for secure V2X communication and digital certificates
- QoS** Quality of Service – Performance metrics such as latency, throughput, and reliability
- RL** Reinforcement Learning – Machine learning approach where agents learn optimal actions through interaction
- RSU** Road Side Unit – Infrastructure equipment that communicates with vehicles for V2I applications
- SoC** System on Chip – Integrated circuit combining CPU, memory, peripherals, and wireless modules
- UART** Universal Asynchronous Receiver-Transmitter – Serial communication protocol used between ESP32 and Jetson Nano
- URLLC** Ultra-Reliable Low-Latency Communication – 5G service category for mission-critical applications
- Uu** Network Interface – Cellular interface in C-V2X connecting vehicles to mobile network infrastructure
- V2G** Vehicle-to-Grid – Energy exchange between electric vehicles and the power grid (charging and discharging)
- V2I** Vehicle-to-Infrastructure – Communication between vehicles and roadside units, traffic lights, or other infrastructure
- V2N** Vehicle-to-Network – Communication with a cellular or backend server (cloud) for telematics, remote monitoring, and control
- V2P** Vehicle-to-Pedestrian – Safety alerts and communication between vehicles and vulnerable road users (pedestrians, cyclists)
- V2V** Vehicle-to-Vehicle – Communication between vehicles to share position, speed, and safety messages
- V2X** Vehicle-to-Everything – Umbrella term covering all communication types between a vehicle and its environment

Abstract

The rapid growth of electric vehicles (EVs) and intelligent transportation systems has intensified the need for advanced Vehicle-to-Everything (V2X) communication technologies. These technologies are expected to improve road safety, enable energy management, and support the transition toward smart and sustainable mobility. However, the deployment of V2X remains challenging, particularly due to the lack of accessible prototyping platforms, limited interoperability, and the absence of low-cost solutions suitable for academic and experimental research.

This project addresses these gaps by developing a custom V2X prototyping platform based on the Renault Twizy electric vehicle. A low-cost On-Board Unit (OBU) was designed around an NVIDIA Jetson Nano and ESP32 microcontrollers to acquire, process, and transmit real-time CAN bus data across different wireless communication protocols.

Two V2X subdomains were prototyped. First, a Vehicle-to-Vehicle (V2V) proof of concept was implemented using ESP-NOW to broadcast safety-critical data such as speed, gear state, and battery State of Charge (SoC). Second, a Vehicle-to-Grid (V2G) prototype was developed through the design of a custom charging station, enabling remote monitoring and session management via a ThingsBoard dashboard.

Despite the inability to achieve actual bidirectional power transfer due to hardware and safety constraints, the system successfully demonstrated bidirectional information exchange, real-time data monitoring, and integration with a local IoT platform. These results confirm the feasibility of the proposed low-cost architecture and provide a foundation for future enhancements, including ISO 15118-compliant communication, certified bidirectional chargers, large-scale multi-vehicle testing, and novel applications such as healthcare-oriented V2G for home resilience.

General Introduction

General Context

In recent years, the world has witnessed major advancements in sectors such as wireless communication, renewable energy, power management, semiconductors, and artificial intelligence. These innovations have profoundly impacted the automotive industry, leading to the mass production of electric vehicles (EVs), as shown in Fig. 1, and the development of advanced mobility systems. With new vehicles like the Tesla Model 3, Ford Mustang Mach-E, and Kia EV6 already on the market, the shift toward connected and autonomous mobility has become a tangible reality [16]. It is not an exaggeration to state that the automotive sector is currently undergoing its most significant transformation since Henry Ford industrialized car manufacturing a century ago. Modern vehicles are increasingly embedded with highly advanced technologies, including sensors such as Lidars, cameras, onboard computers, and algorithms that enhance their awareness of their surroundings and grant them partial driving control. This evolution highlights the growing importance of Vehicle-to-Everything (**V2X**) technologies [22], which unlock the full potential of next-generation EVs by enabling seamless interaction with other vehicles (V2V), the power grid (V2G), infrastructure (V2I), and even pedestrians (V2P).

This project explores the practical implementation of these V2X concepts on a specific electric vehicle platform: the Renault Twizy. The Twizy represents a class of compact, urban-focused EVs whose unique characteristics—such as its small footprint and agility—make it an ideal candidate to benefit from V2X connectivity in dense city environments. Applications like collision warning at intersections and receiving traffic light information are particularly relevant to its use case.

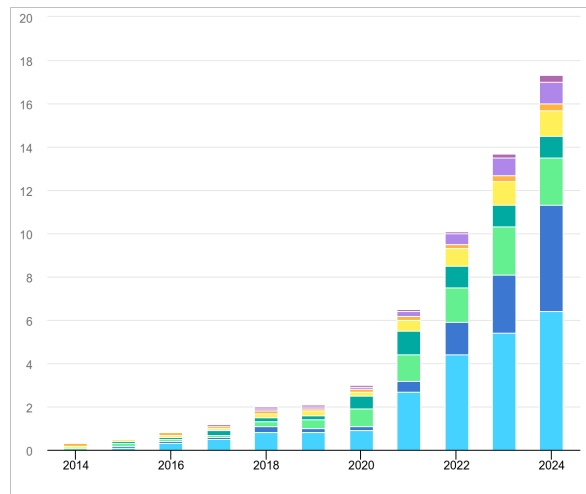


Figure 1: Global electric car sales, 2014-2024 (IEA). The rapid growth underscores the industry shift towards electrification.

Source: International Energy Agency (IEA), CC BY 4.0

Motivations and Problem Statement

The rapid adoption of electric vehicles (EVs) and the emergence of connected and autonomous mobility create new opportunities but also introduce challenges that must be addressed before large-scale deployment can be achieved. Although Vehicle-to-Everything (V2X) technologies are a key enabler of safer, smarter, and more efficient mobility, their real-world implementation faces several obstacles:

- **High Cost of Commercial Solutions:** Existing V2X communication units often rely on DSRC or C-V2X chipsets and certified automotive-grade hardware, which are expensive and difficult to source for research or prototyping purposes.
- **Complexity of Integration:** Many EVs, including compact urban vehicles such as the Renault Twizy, are not equipped with native V2X capabilities. Integrating new communication systems into their electronic architecture typically requires specialized knowledge of CAN protocols, vehicle diagnostics, and embedded systems.
- **Limited Experimental Platforms:** Most available research testbeds focus on large vehicles or controlled laboratory environments, leaving a gap for affordable, small-scale platforms that can be used by students, researchers, or startups to test V2X use cases in real urban settings.
- **Lack of Open, Flexible Prototypes:** There is a shortage of open and customizable V2X development kits that allow rapid iteration, experimentation, and validation of V2V, V2I, and V2G applications without the constraints of industry-standard compliance at early research stages.

Given these challenges, there is a clear need for a low-cost, modular, and flexible V2X prototyping solution that can be quickly deployed on small EV platforms. Such a system would enable researchers and developers to explore use cases such as collision warnings, cooperative driving, and smart-charging coordination, while avoiding the high costs and complexity of commercial automotive systems.

This project addresses this gap by developing a custom, low-cost On-Board Unit (OBU) based on readily available hardware (Jetson Nano, ESP32), and by integrating it with the Renault Twizy’s CAN bus to create a practical and extensible V2X experimentation platform.

Project Objectives and Goals

This internship project focused on the design, implementation, and testing of a custom V2X prototyping system on a Renault Twizy electric vehicle, with an emphasis on V2V and V2G functionalities. As a proof-of-concept and a first exploration into automotive systems, this project prioritized accessibility and core functionality over compliance with standard automotive protocols (e.g., DSRC, C-V2X, LTE-V2X). Within this technical framework, the core objectives of the project were as follows:

- **To design and develop a custom, low-cost On-Board Unit (OBU)** based on an NVIDIA Jetson Nano. This OBU was tasked with interfacing with the vehicle’s Controller Area Network (CAN bus) to read vehicle data, processing this data, and managing both short-range (ESP-NOW) and long-range (Wi-Fi) communication.
- **Establish and demonstrate basic V2V communication** using the ESP-NOW protocol. The objective was to enable the Twizy to broadcast its core data (e.g., position, speed, heading) and receive similar data from other equipped vehicles to simulate standard Basic Safety Messages (BSMs).
- **Implement a V2G proof-of-concept** focusing on bidirectional information exchange (charging status, control signals), while acknowledging that actual bidirectional power transfer was outside the project scope due to hardware and safety constraints.
- **Implement a V2N (Vehicle-to-Network) system** using the Jetson Nano’s Wi-Fi connectivity. The goal was to transmit collected vehicle data to a centralized enabling remote monitoring and data logging.

By meeting these objectives, the project delivered a functional V2X research platform built with accessible technology, demonstrating the feasibility of core connected-vehicle concepts and laying the groundwork for future extensions toward standardized and industry-compliant systems.

Host Organization – University UIR and LERMA Laboratory

Introduction

The International University of Rabat (UIR) is a prominent educational institution in Morocco, dedicated to merging tradition with modernity and fostering an environment of innovation and intellectual advancement. Established in 2009, UIR has made significant strides in higher education and research, positioning itself as a leader in scientific inquiry and technological development. Central to UIR's mission is its commitment to providing cutting-edge education and promoting groundbreaking research, aimed at shaping future leaders who will contribute significantly to Morocco's intellectual and social landscape.

This chapter explores the foundational aspects of UIR, its academic and research excellence, and its substantial contributions to science and technology. It also highlights the pivotal role of the LERMA Laboratory within UIR's research infrastructure, focusing on its research themes, expertise, key projects, and achievements. Furthermore, the chapter presents the strategic importance of the Energy-Efficient Building (EEB) Platform, its innovative technologies and methodologies, and its contributions to sustainable development. Finally, it reflects on UIR's impact on the research landscape and outlines future projects that emphasize UIR's commitment to excellence and innovation.

History and Mission

The International University of Rabat (UIR) stands as a leading educational institution in Morocco, envisioned by Professor Nouredine Mouaddib in 2006. Officially established in 2009, UIR signed a development contract with the Moroccan government, reaffirming its dedication to academic excellence. The foundation of UIR was symbolically marked in 2010 by King Mohammed VI, who laid the first stone of the campus, thus initiating a journey toward innovation and intellectual advancement.

UIR's mission is deeply rooted in delivering cutting-edge higher education and promoting groundbreaking research. The university strives to merge tradition with modernity, shaping future leaders who will play significant roles in Morocco's intellectual and social spheres. UIR's vision emphasizes adherence to global educational standards while maintaining a strong connection to local cultural and social contexts.

Academic and Research Excellence

UIR is acclaimed for its academic rigor and research excellence. The university offers a diverse range of programs designed to meet the evolving demands of various industries. These programs are supported by advanced research initiatives that contribute substantially to science and technology. UIR is home to multiple centers of excellence, each dedicated to advancing knowledge and promoting innovation in specialized fields. These centers play a crucial role in fostering intellectual growth and reinforcing UIR's reputation as a hub for academic and research excellence.



Figure 2: Research laboratories and centers at UIR.

Contributions to Science and Technology

The contributions of UIR to science and technology are substantial. Through its diverse research programs and collaborations, the university has positioned itself as a leader in scientific inquiry and technological advancement. UIR's research outputs are frequently published in renowned journals and translated into practical applications that benefit society.

LERMA Laboratory

Role and Significance



Figure 3: LERMA Laboratory at UIR.

The LERMA Laboratory is a cornerstone of UIR's research infrastructure. It plays a crucial role in enhancing the university's research capabilities and academic reputation. LERMA's contributions extend beyond UIR, impacting the wider scientific community through its innovative research and development initiatives.

Key Projects and Achievements

LERMA has undertaken numerous high-impact projects that have gained significant recognition. These projects range from developing renewable energy technologies to creating sustainable solutions for urban development. The laboratory's achievements demonstrate its commitment to excellence and its capacity to translate research into tangible societal benefits.

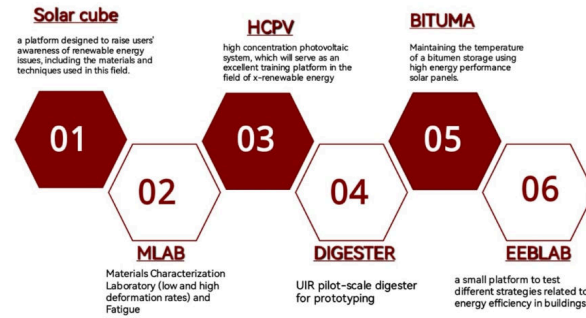


Figure 4: Key projects carried out at LERMA Laboratory.

Research Themes and Expertise

LERMA's research focuses on the design and development of smart materials, structures, and systems used in aeronautical, automotive, and energy technologies. It primarily hosts research activities conducted by faculty and scientists from four engineering schools: AERO, AUTO, ESIE, and ESIN. Research is directed toward developing methods and tools for designing large-scale systems with the ability to learn, adapt, and self-organize. These systems continuously evolve, redesigning their structure and behavior to suit their environment.

Recent advances in miniaturized microsystems have made it possible to develop systems capable of sensing their environment and reacting accordingly. Research in smart materials is progressing toward adaptive systems with evolvable operational capabilities. These advances—encompassing smart materials, nanotechnologies, communication systems, and AI—enable the development of intelligent systems that can autonomously respond to environmental changes. For instance, smart building systems can react in real-time to factors such as occupancy, weather, and load conditions.

The Energy-Efficient Building (EEB) Platform

Purpose and Strategic Importance

The Energy-Efficient Building (EEB) Platform, also known as EEBLAB, is a strategic initiative by UIR aimed at promoting sustainable building practices. Its primary goal is to develop and implement energy-efficient technologies and methodologies that can be applied to modern construction projects. This initiative plays a crucial role in addressing the environmental challenges posed by rapid urbanization and industrialization.



Figure 5: The Energy-Efficient Building (EEB) Platform at UIR.

Contributions to Sustainable Solutions

The EEB Platform contributes significantly to sustainable development by promoting energy-efficient building practices. Its efforts reduce the environmental impact of construction and foster greener, more sustainable urban environments. The platform's innovations have the potential to revolutionize the construction industry, making it more environmentally responsible. Publications and reports detailing these contributions are available on the official EEBLAB website: <https://eeb.uir.ac.ma/>.

Technologies and Methodologies

The EEB Platform utilizes innovative technologies and research methodologies, including advanced insulation materials, energy-efficient HVAC systems, and smart building technologies that optimize energy use. Its research approach ensures that the developed solutions are both practical and scalable.

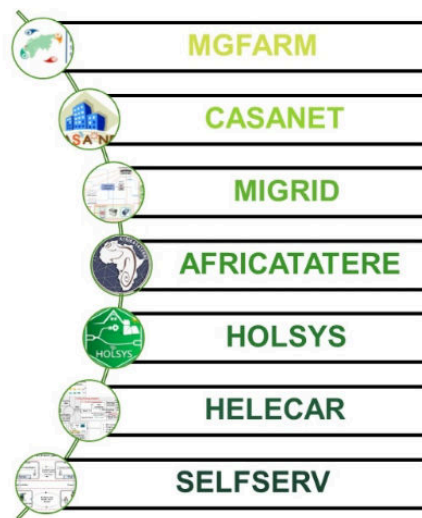


Figure 6: Key projects under the EEBLAB initiative.

Impact and Future Prospects

Influence on the Research Landscape

Through its research initiatives and academic programs, UIR has made a profound impact on the research landscape nationally and internationally. Its contributions are widely recognized in scholarly publications and have shaped research directions in several key areas.

Future Directions and Projects

Looking ahead, UIR is committed to advancing its research capabilities and expanding its academic offerings. Several ambitious projects are planned, reflecting UIR's strategy to foster a dynamic research environment that addresses societal needs.

Conclusion

The International University of Rabat (UIR) has established itself as a beacon of academic and research excellence in Morocco. Through comprehensive academic programs and innovative research initiatives, UIR continues to make significant contributions to science and technology at both national and international levels. The LERMA Laboratory exemplifies UIR's dedication to innovation, driving forward the development of smart materials, systems, and technologies that address key challenges in multiple sectors.

The Energy-Efficient Building (EEB) Platform further highlights UIR's commitment to sustainability and environmental stewardship. By promoting energy-efficient practices and developing scalable, practical solutions, EEBLAB contributes to sustainable urban development.

As UIR looks toward the future, it remains steadfast in its mission to blend tradition with modernity, shaping future leaders and advancing the boundaries of science and technology. With ambitious projects on the horizon, UIR is well-positioned to continue its trajectory of excellence and contribute to a smarter, more connected, and sustainable world.

State of the Art

1.1 Vehicle-to-Everything Fundamentals

1.1.1 Definition, Benefits, and Current Trends

Vehicle-to-Everything (V2X) represents a transformative communication paradigm that enables vehicles to exchange information with their entire surrounding ecosystem, including other vehicles (V2V), infrastructure (V2I), networks (V2N), pedestrians (V2P), and the electric grid (V2G). This comprehensive connectivity framework aims to revolutionize transportation by enhancing road safety, optimizing traffic flow, reducing environmental impact, and enabling advanced autonomous driving capabilities while supporting new energy management services.

Recent empirical studies demonstrate the substantial quantitative benefits of V2X deployment. A comprehensive 2025 comparative analysis of C-V2X versus DSRC in congested urban environments revealed that C-V2X achieves latency reductions exceeding 90% compared to DSRC while simultaneously decreasing traffic conflict rates by approximately 38% when implemented at high penetration rates [18]. Furthermore, extensive survey research conducted in 2025 indicates that V2X applications including cooperative perception, intersection movement assistance, and emergency vehicle prioritization are transitioning from laboratory simulations to real-world pilot deployments across Europe, Asia, and North America, with particularly promising results in smart city initiatives [10].

The evolution of V2X technology is increasingly driven by the convergence of multiple technological domains. Machine learning algorithms integrated with advanced sensor fusion techniques enable vehicles to create comprehensive situational awareness that extends beyond individual sensor limitations. Recent developments in edge computing architectures have demonstrated significant improvements in real-time processing capabilities, reducing computational latency from hundreds of milliseconds to sub-10ms response times for critical safety applications [20]. Additionally, the integration of digital twin technologies allows for real-time synchronization between physical vehicles and their virtual counterparts, enabling predictive maintenance, route optimization, and enhanced fleet management capabilities [25].

1.2 Communication Technologies and Standards

1.2.1 Radio Access Technologies: DSRC vs C-V2X Evolution

The V2X communication landscape is characterized by two primary radio access technologies, each offering distinct advantages and facing unique challenges. Dedicated Short-Range Communications (DSRC) based on IEEE 802.11p and its European variant ITS-G5 represent the mature, extensively tested foundation of vehicle safety communications. DSRC systems offer inherently low latency communication with simplified hardware requirements, making them particularly suitable for time-critical safety applications such as collision avoidance and emergency braking assistance. However, DSRC faces significant limitations including spectrum constraints, limited infrastructure deployment scalability, and reduced reliability in non-line-of-sight environments, particularly in dense urban scenarios

with high-rise buildings and complex road geometries.

Cellular Vehicle-to-Everything (C-V2X) technology, built upon LTE and evolving toward 5G New Radio (NR-V2X), provides both direct communication modes (PC5 interface) and network-assisted communication (Uu interface). This dual-mode capability enables C-V2X to offer greater communication range, improved scalability, and seamless integration with existing cellular infrastructure. Recent performance evaluations demonstrate that C-V2X systems can maintain reliable communication links at distances exceeding 1 kilometer in highway scenarios, compared to DSRC's typical 300-500 meter range [13]. The integration of 5G capabilities introduces ultra-reliable low-latency communication (URLLC) features that support mission-critical applications with latency requirements below 1 millisecond.

adjustbox

V2X Technology	Dedicated Short Range Communications (DSRC)	Cellular Vehicle-to-Everything (C-V2X)
Technical Development	Early development, more mature	Maturity expected between 2020 and 2022
Vehicle Manufacturers	GM, JLR, Toyota, Volkswagen	Audi, BMW Group, Ford, Geely, Honda, Lexus, Mercedes Benz, Daimler, PSA Group
Chip Suppliers	NXP, Renesas, Qualcomm, Savari	Intel, Qualcomm, Huawei, Quectel, Keysight Technologies, LG Innotek
Frequency band	5.8 – 5.9 MHz in Europe, US and Japan	Compatible with existing 4G and future 5G
Network connection capability	• Supports up to 200 Km/hr • Response time < 100 ms • Data transmission > 12 Mbps • Transmission range < 1 km	• Supports up to 500 km/hr • Response time < 50 ms • Data transmission < 500 Mbps • Transmission range < 1 km
Upsides	• Mature standards and technology • Products available in Europe, US and Japan	• Works on existing 4G (LTE) network • Large bandwidth, good synchronization and long transmission distance
Downsides	• Not compatible with existing 4G base stations • Local roadside base stations needed	• Standards are being developed • Little market experience

Table 1.1: Comparison of V2X Technologies: DSRC and C-V2X

Recent research on spectrum coexistence has addressed one of the most pressing challenges in V2X deployment. A comprehensive study examining DSRC and C-V2X coexistence within the 5.895-5.925 GHz ITS band found that optimal channel allocation strategies vary significantly based on deployment scenarios [27]. In highway environments, allocating two channels to DSRC and one to C-V2X provides optimal performance balance, while urban scenarios benefit from the reverse allocation due to different interference characteristics and propagation patterns. These findings are particularly relevant for countries implementing hybrid V2X strategies and for experimental systems requiring interoperability between different technologies.

Therefore, countries around the world are combining their strategies for 5G IoV (Internet of Vehicle) technology. In addition to taking large steps toward the era of self-driving, they are enthusiastic about

the development of related industries and supply chains, and even general consumers are interested in the widespread commercialization. The following describes the IoV strategies of several major regions, starting with the allocation of spectra, as shown in the following table:

Table 1.2: Allocation of Spectra in Major Regions

Country	V2X Technology	Frequency band used	Bandwidth	Frequency band sections
USA	DSRC / C-V2X	5905–5925 MHz	20 MHz	Ch182–184: Dedicated to C-V2X
		5895–5905 MHz	10 MHz	Ch180: DSRC/C-V2X (FCC 2020)
		–	–	DSRC phase-out after transition (FCC 2023)
EU	ITS-G5 / C-V2X	5470–5725 MHz	255 MHz	Permit-free band
		5795–5815 MHz	20 MHz	Vehicle-to-road system
		5855–5875 MHz	20 MHz	Non-safety ITS applications
		5875–5925 MHz	50 MHz	Safety ITS applications
		63–64 GHz	1 GHz	Advanced ITS applications
China	C-V2X	5905–5925 MHz	20 MHz	Dedicated to C-V2X

1.2.2 Standardization and Regulatory Framework

Global standardization efforts continue to evolve rapidly, with significant regulatory developments shaping the V2X landscape. The Federal Communications Commission (FCC) in the United States has implemented comprehensive rules for 5.9 GHz band allocation, officially designating specific spectrum portions for C-V2X deployments while maintaining coexistence provisions for existing DSRC systems [3]. Similarly, the European Telecommunications Standards Institute (ETSI) and the 3rd Generation Partnership Project (3GPP) continue collaborative work on NR-V2X enhancements, focusing on advanced features such as sidelink communication improvements, enhanced positioning accuracy, and support for high-mobility scenarios with vehicles traveling at speeds exceeding 250 km/h [4].

The International Organization for Standardization (ISO) has made significant progress in standardizing bidirectional communication protocols through ISO 15118, which defines the communication interface between electric vehicles and charging infrastructure. This standard has been extended to support Vehicle-to-Grid (V2G) applications, enabling standardized energy exchange protocols that are essential for grid integration and energy trading applications [7].

1.3 Advanced V2X Application Domains

1.3.1 Safety-Critical and Traffic Efficiency Applications

Contemporary V2X safety applications extend far beyond basic collision warnings to encompass sophisticated cooperative systems that leverage multiple data sources for enhanced situational awareness. Cooperative perception represents one of the most promising advancement areas, where vehicles and infrastructure components share sensor data to create comprehensive environmental maps that extend individual sensor capabilities. This technology enables vehicles to "see around corners" and detect obstacles or hazards that would otherwise remain hidden behind buildings, other vehicles, or terrain features. Recent research by Huang et al. demonstrates that cooperative perception systems can improve hazard detection range by up to 200% compared to individual vehicle sensors, while reducing false positive rates through data fusion algorithms that correlate information from multiple sources [6].

Advanced intersection management systems represent another critical application domain where V2X technology demonstrates substantial benefits. These systems coordinate vehicle movements through complex intersections by optimizing traffic light timing, providing priority to emergency vehicles, and managing pedestrian crossing phases. Real-world deployments in European smart cities have shown traffic throughput improvements of 25-30% during peak hours, while simultaneously reducing average waiting times and fuel consumption [5]. Weather-responsive traffic management systems utilize V2X communication to disseminate real-time weather conditions, road surface information, and visibility data, enabling dynamic speed limit adjustments and route recommendations that improve safety during adverse conditions.

1.3.2 Vehicle-to-Grid (V2G) and Energy Management

Vehicle-to-Grid technology represents a paradigm shift in energy infrastructure, transforming electric vehicles from simple energy consumers into active participants in grid stabilization and energy distribution. V2G systems enable bidirectional power flow between vehicles and the electrical grid, allowing parked electric vehicles to serve as distributed energy storage systems that can provide grid services such as frequency regulation, voltage support, and peak load shaving. Recent implementations demonstrate that a fleet of 1000 connected electric vehicles can provide equivalent grid stabilization services to a 10 MW conventional power plant, while offering significantly faster response times due to the instantaneous nature of electronic power conversion [8].

Healthcare applications of V2G technology have emerged as particularly critical use cases, especially following lessons learned from recent natural disasters and grid reliability challenges. Hospitals and medical facilities equipped with V2G-enabled vehicle fleets can maintain critical power supplies during grid outages, with electric ambulances and medical transport vehicles serving as mobile power sources for essential medical equipment. A pilot program in California demonstrated that a fleet of 50 electric ambulances could provide backup power for a regional medical center for up to 72 hours during emergency situations, while maintaining sufficient range for emergency response operations [12]. Emergency response applications extend this concept further, with disaster relief organizations utilizing V2G-enabled vehicles to establish temporary power infrastructure in affected areas, providing electricity for communication systems, medical equipment, and essential services.

1.3.3 Smart City Integration and Mobility Services

The integration of V2X technology with smart city infrastructure creates synergistic effects that enhance urban mobility, environmental sustainability, and quality of life. Advanced traffic management systems leverage V2X communication to implement dynamic routing strategies that consider real-time traffic conditions, parking availability, air quality measurements, and energy consumption optimization. These systems can automatically adjust traffic signal timing, implement congestion pricing, and coordinate public transportation schedules to minimize overall travel time and environmental impact [1].

Shared mobility services benefit significantly from V2X connectivity through enhanced fleet management, predictive maintenance, and optimized vehicle utilization. Car-sharing and ride-hailing platforms utilize V2X data to monitor vehicle health in real-time, predict maintenance requirements, and optimize fleet positioning to reduce user waiting times. Recent studies indicate that V2X-enabled shared mobility fleets achieve 40% higher utilization rates compared to conventional systems, while reducing maintenance costs through predictive analytics that identify potential issues before they result in vehicle breakdowns [2].

1.4 Security, Privacy, and Trust Frameworks

1.4.1 Cybersecurity Challenges and Solutions

The increasing connectivity of vehicles introduces significant cybersecurity challenges that must be addressed to ensure safe and trustworthy V2X deployment. Modern V2X security frameworks implement multi-layered protection strategies that include cryptographic authentication, message integrity verification, and misbehavior detection systems. Public Key Infrastructure (PKI) systems specifically

designed for V2X applications provide the foundation for secure communication by managing digital certificates, enabling pseudonym changes for privacy protection, and facilitating trust establishment between communicating entities [28].

Advanced misbehavior detection systems utilize machine learning algorithms to identify and isolate vehicles transmitting false, corrupted, or malicious information. These systems analyze communication patterns, message consistency, and behavioral anomalies to detect potential security threats in real-time. Recent developments include distributed detection mechanisms that leverage collective intelligence from multiple vehicles to improve detection accuracy while reducing computational overhead on individual vehicles [9].

1.4.2 Privacy Protection and Data Management

Privacy preservation in V2X systems requires sophisticated approaches that balance safety requirements with individual privacy rights. Contemporary privacy frameworks implement dynamic pseudonym management systems that regularly change vehicle identifiers to prevent long-term tracking while maintaining short-term communication continuity necessary for safety applications. Recent research introduces privacy-preserving cooperative perception systems that enable vehicles to share sensor data without revealing specific location information or movement patterns [14]. These systems utilize advanced cryptographic techniques such as homomorphic encryption and secure multi-party computation to enable collaborative data processing while protecting individual privacy.

Location privacy represents a particular challenge in V2X systems, as safety applications require precise position information while users demand protection against surveillance and tracking. Novel approaches include differential privacy techniques that add controlled noise to location data, k-anonymity methods that group vehicles with similar characteristics, and temporal cloaking mechanisms that delay message transmission to prevent real-time tracking [19].

1.5 Emerging Technologies and Future Directions

1.5.1 Artificial Intelligence and Machine Learning Integration

The integration of artificial intelligence and machine learning technologies with V2X systems enables adaptive, intelligent transportation solutions that continuously improve performance through experience and data analysis. Advanced AI algorithms process vast amounts of V2X data to identify traffic patterns, predict congestion, and optimize routing decisions in real-time. Deep learning models trained on V2X communication data can predict vehicle behavior, estimate arrival times, and detect anomalous situations that may indicate accidents, breakdowns, or security threats [24].

Reinforcement learning applications in V2X systems enable autonomous adaptation to changing traffic conditions and user preferences. These systems learn optimal strategies for traffic light control, route recommendation, and energy management through continuous interaction with the transportation environment. Recent implementations demonstrate that RL-based traffic control systems can reduce average travel time by up to 20% while improving energy efficiency and reducing emissions [15].

1.5.2 6G and Next-Generation Connectivity

Research into 6G wireless technology promises revolutionary improvements in V2X communication capabilities, including support for holographic communication, brain-computer interfaces for vehicles, and integration with satellite networks for global coverage. 6G systems are expected to provide communication latency below 100 microseconds, enable precise positioning with centimeter-level accuracy, and support massive IoT deployments with millions of connected devices per square kilometer [11]. These capabilities will enable new applications such as real-time haptic feedback for remote vehicle operation, augmented reality navigation systems, and advanced cooperative autonomous driving scenarios.

1.6 Research Gaps and Implementation Challenges

Despite significant advances in V2X technology, several critical research gaps and implementation challenges remain. Interoperability between different V2X standards and technologies continues to pose deployment challenges, particularly in regions where multiple standards coexist. Performance evaluation under mixed traffic conditions, where connected and non-connected vehicles share the same infrastructure, requires further investigation to understand real-world deployment scenarios. Economic models for V2X services, particularly V2G energy trading and cooperative perception data sharing, need development to enable sustainable business models and widespread adoption.

Scalability concerns arise when considering city-wide or national-level V2X deployment, as current systems have primarily been tested in controlled environments or limited pilot programs. The integration of V2X systems with existing traffic management infrastructure presents technical and regulatory challenges that vary significantly across different regions and jurisdictions. Additionally, public acceptance and trust in V2X technology remain critical factors that influence adoption rates and require ongoing attention to privacy, security, and reliability concerns [21].

System Requirements & Design

2.1 Project Requirements

For realizing this V2X communication project, we required essential hardware and software equipment, tools, devices, and components to implement (V2V) and (V2G) communication systems. The core requirements include an On-Board Unit (OBU) for the vehicle, a Renault Twizy electric vehicle as the test platform, an Electric Vehicle Charging Station (EVCS) with Road-Side Unit (RSU) capabilities, and a cloud-based server infrastructure for data management and visualization.

2.2 Hardware Components Overview

Before diving into the detailed system architecture, it is essential to understand the key hardware components that form the foundation of our V2X communication system. This section provides an overview of the primary components used in both the vehicle-mounted OBU and the infrastructure-based RSU.

2.2.1 Microcontrollers and Computing Platforms

ESP32 Microcontroller

The ESP32 is a low-cost, low-power system-on-chip (SoC) microcontroller with integrated Wi-Fi and Bluetooth capabilities. It features a dual-core Tensilica Xtensa LX6 microprocessor, making it suitable for real-time applications requiring concurrent processing tasks. In our project, the ESP32 serves as the primary communication interface for V2V and V2G protocols, handling both ESP-NOW peer-to-peer communication and traditional Wi-Fi connectivity.

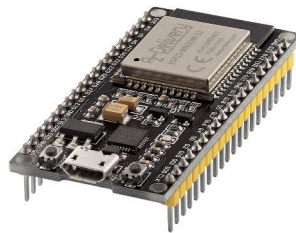


Figure 2.1: ESP32 microcontroller development board

Key specifications include 520 KB of SRAM, up to 16 MB of external flash memory, and operating voltage range of 2.2V to 3.6V. The integrated wireless capabilities eliminate the need for external communication modules, reducing system complexity and cost.

NVIDIA Jetson Nano

The NVIDIA Jetson Nano is a small, powerful computer specifically designed for AI and machine learning applications at the edge. It features a 128-core Maxwell GPU and quad-core ARM A57 CPU, providing significant computational power for processing vehicle telemetry data, running machine learning inference models, and managing complex communication protocols.

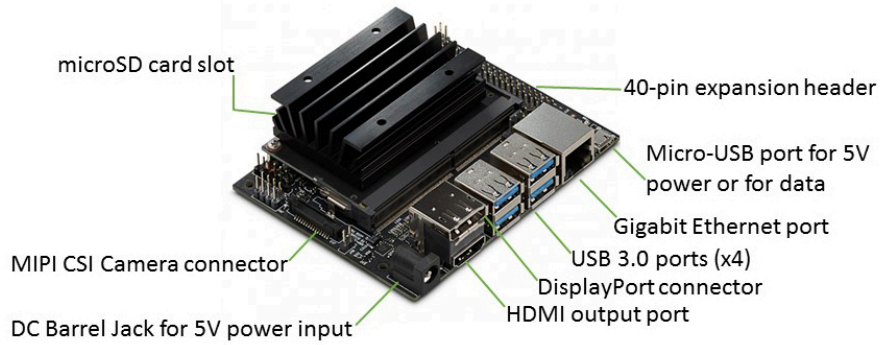


Figure 2.2: NVIDIA Jetson Nano development board

In our OBU design, the Jetson Nano acts as the central processing unit, responsible for receiving and decoding CAN bus data from the ESP32, executing V2X application logic, and managing data transmission to cloud services. Its Linux-based operating system enables the use of standard networking protocols and development tools.

Raspberry Pi 3

The Raspberry Pi 3 is a credit-card-sized single-board computer running a full Linux operating system. It features a quad-core ARM Cortex-A53 processor, 1GB of RAM, and integrated Wi-Fi and Bluetooth connectivity. The Raspberry Pi provides a cost-effective platform for implementing RSU functionality at charging stations.

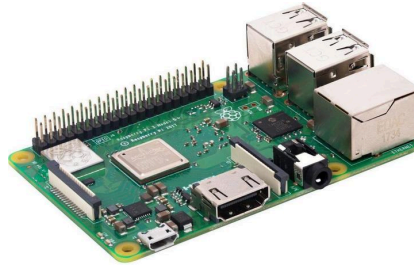


Figure 2.3: Raspberry Pi 3 single-board computer

In the EVCS RSU, the Raspberry Pi manages data collection from power measurement sensors, processes V2G communication protocols, and publishes telemetry data to cloud-based monitoring systems using MQTT.

2.2.2 Communication and Interface Modules

MCP2515 CAN Controller

The MCP2515 is a standalone Controller Area Network (CAN) controller that implements the CAN 2.0B protocol. When paired with a CAN transceiver such as the TJA1050, it provides a complete interface solution for accessing vehicle CAN bus networks through standard SPI communication.

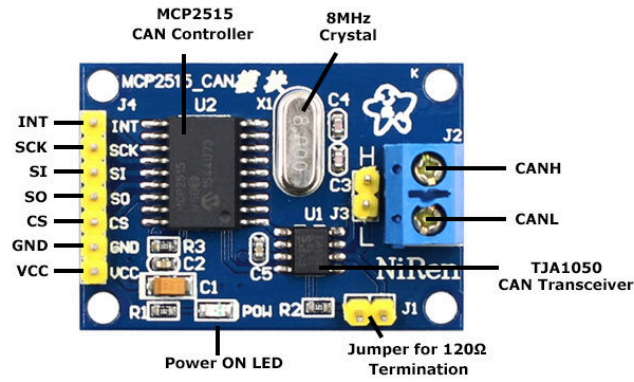


Figure 2.4: MCP2515 CAN controller module with TJA1050 transceiver

This module enables our ESP32 to read real-time vehicle data including speed, battery state of charge, voltage, and current from the Twizy’s internal CAN network via the OBD-II diagnostic port.

PZEM-004T Power Measurement Module

The PZEM-004T is a comprehensive power measurement module capable of monitoring voltage, current, power, and energy consumption in AC electrical systems. It communicates via UART protocol and provides high accuracy measurements suitable for V2G power management applications.



Figure 2.5: PZEM-004T power measurement module

In our EVCS implementation, the PZEM-004T monitors the electrical parameters of the charging process, enabling real-time power flow analysis and grid integration monitoring for V2G applications.

2.2.3 Network Infrastructure Components

TP-Link M7200 Mobile Hotspot

The TP-Link M7200 is a portable 4G LTE mobile Wi-Fi hotspot that provides internet connectivity for V2N (Vehicle-to-Network) communication when fixed network infrastructure is unavailable. It supports up to 10 simultaneous connections and offers download speeds up to 150 Mbps.



Figure 2.6: TP-Link M7200 4G LTE mobile hotspot

This device enables the Jetson Nano in our OBU to maintain continuous connectivity to cloud services for telemetry data transmission and remote monitoring, even when the vehicle is operating in areas without fixed Wi-Fi infrastructure.

2.3 Test Vehicle Platform

2.3.1 Renault Twizy Electric Vehicle



Figure 2.7: Renault Twizy electric vehicle

The Renault Twizy serves as our primary test platform for V2X communication development. This compact, fully electric urban vehicle is classified as a heavy quadricycle (L7e category) and offers several advantages for research and prototyping applications. Its lightweight design, small footprint, and simplified architecture make it an excellent platform for prototyping connected vehicle applications such as V2X communication systems.

Key Vehicle Characteristics

The Twizy's electric powertrain consists of a 6.8 kWh lithium-ion battery pack operating within a nominal voltage range of 46-58V, making it safer and more manageable for experimental setups compared to high-voltage electric vehicles. The compact dimensions of 2.33m in length and 1.24m in width provide high maneuverability and make it ideal for urban testing scenarios where space constraints are a consideration.

The vehicle features a unique tandem seating arrangement where the driver and passenger sit one behind the other, optimizing the vehicle footprint while maintaining functionality. The chassis, developed by Renault Sport, provides stability and consistent roadholding characteristics that are essential for repeatable test runs and reliable data collection during V2X experiments.

One of the most significant advantages for our research purposes is the Twizy's accessible Controller Area Network (CAN bus) through its standard OBD-II diagnostic connector. This accessibility enables real-time monitoring of critical vehicle parameters including speed, battery state of charge (SOC), voltage, current levels, and diagnostic fault codes, all of which are essential for comprehensive V2X experimentation and validation.

Performance Specifications and Operating Range

The Renault Twizy is available in two performance variants: a limited 45 km/h version with 5 horsepower and an unrestricted 80 km/h version with 17 horsepower. For this project, we utilized the 80 km/h variant as it provides more realistic road-testing conditions and better represents typical urban driving scenarios where V2X communication would be deployed.

The traction battery provides an operational range of 40-50 miles (65-80 km) on a full charge under normal driving conditions. Charging is accomplished using a standard household electrical socket (230V AC), requiring approximately 3.5 hours for a complete charge cycle. This charging simplicity eliminates the need for specialized high-voltage laboratory infrastructure while still enabling meaningful V2G experimentation.

Technical Specifications

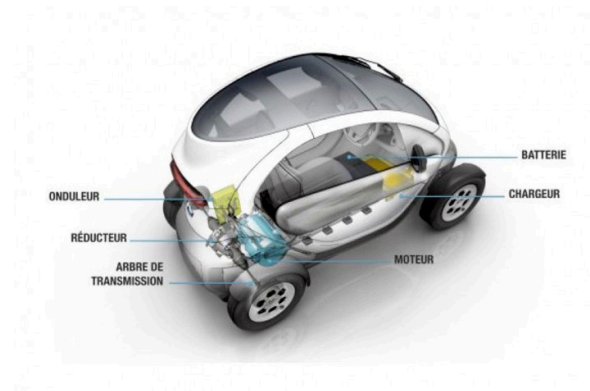


Figure 2.8: Renault Twizy key technical specifications

The vehicle dimensions and technical specifications are shown in Figure 2.9, illustrating the compact form factor that makes the Twizy particularly suitable for urban V2X testing scenarios.

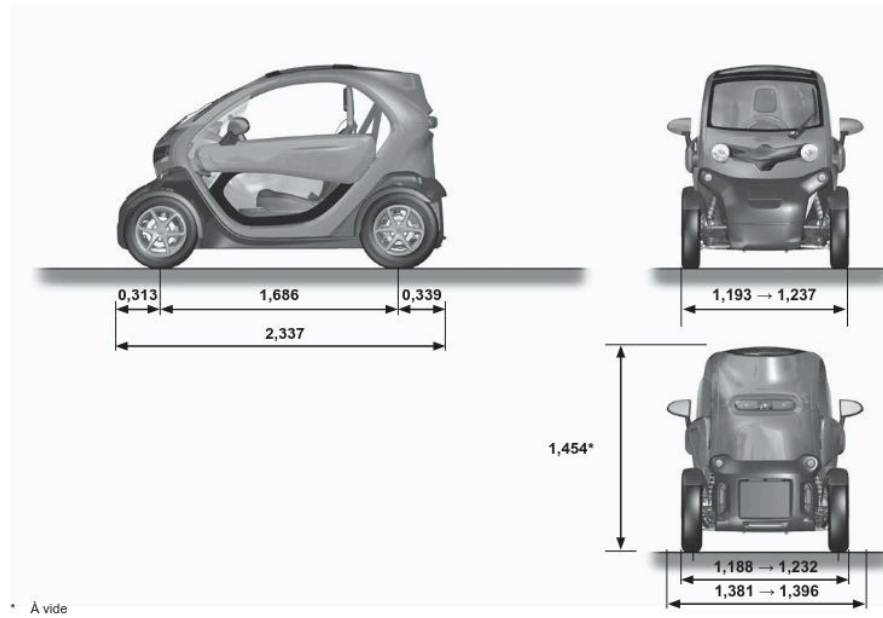


Figure 2.9: Detailed Twizy vehicle dimensions

Powertrain Architecture

The Twizy powertrain consists of integrated components including the traction battery, power inverter, gear motor assembly, and charging system. Understanding this architecture is critical when designing Vehicle-to-Grid (V2G) interfaces and On-Board Unit (OBU) integration points.

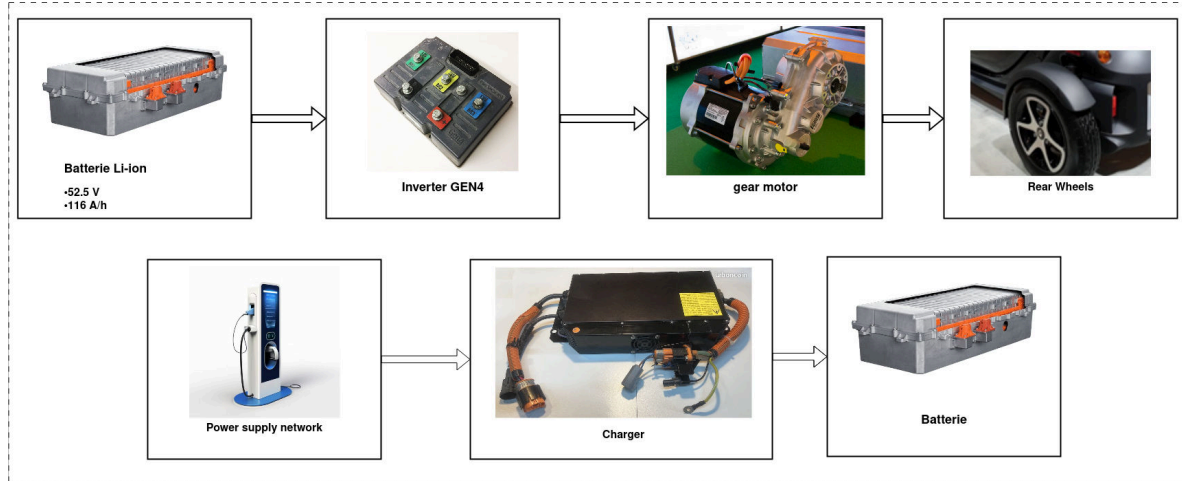


Figure 2.10: Twizy powertrain architecture showing battery, inverter, gear motor, and wheel integration

Dual Battery System Configuration

The Twizy utilizes a dual battery architecture consisting of separate low-voltage and high-voltage systems:

The 12V auxiliary battery is a standard AGM (Absorbed Glass Mat) lead-acid battery that powers all low-voltage vehicle systems including lighting, instrumentation, control electronics, and our custom OBU installation. The specifications are detailed in Table 2.1.

The traction battery provides the main propulsion energy and represents the primary energy storage system for V2G applications. Its moderate voltage level and capacity make it ideal for safe experimental V2G implementations. The complete specifications are presented in Table 2.2.

12V Auxiliary Battery Characteristics	
Voltage	12V
Capacity	12Ah
Battery Type	Lead-acid AGM (waterproof)

Table 2.1: 12V auxiliary battery specifications

Traction Battery Specifications	
Voltage Range	46-58V
Maximum Current	360A
Capacity	6.8kWh
Weight	98kg
Battery Type	Lithium-Ion
Dimensions	828 × 478 × 233mm

Table 2.2: Traction battery technical specifications

Project Relevance and Advantages The Twizy’s accessible CAN bus interface and electric powertrain architecture make it an ideal experimental platform for prototyping both V2V and V2G communication systems. The vehicle’s moderate voltage level (58V maximum) and manageable battery capacity (6.8kWh), combined with its ability to charge directly from standard single-phase 220V AC outlets, significantly simplify the design of safe, cost-effective experimental setups for Vehicle-to-Grid communication research without requiring specialized high-voltage laboratory infrastructure or extensive safety equipment.

2.4 Communication Protocols and Technologies

2.4.1 ESP-NOW for V2V and V2G Communication

ESP-NOW represents a proprietary wireless communication protocol developed by Espressif Systems, specifically designed to enable direct, low-latency, peer-to-peer communication between ESP32 and ESP8266 devices without requiring traditional network infrastructure such as routers or full TCP/IP protocol stacks.

Protocol Characteristics and Advantages

ESP-NOW offers several key features that make it particularly suitable for V2X applications. The protocol typically achieves communication latency under 10 milliseconds, making it well-suited for exchanging time-critical Basic Safety Messages (BSMs) in real-time vehicular communication scenarios. The low power consumption characteristics make it ideal for battery-powered or power-constrained devices, while the infrastructure-independent operation allows vehicles to communicate directly without relying on cellular towers or Wi-Fi access points.

The protocol supports one-to-many communication topologies, enabling multiple peers to broadcast and receive data simultaneously. While ESP-NOW does not implement full multi-hop mesh routing capabilities found in more complex protocols, it effectively supports star and broadcast network topologies that are sufficient for localized group communication scenarios typical in V2V applications.

Technical Implementation Using Action Frames

ESP-NOW leverages the IEEE 802.11 Wi-Fi standard’s management frame structure to transmit data efficiently. In Wi-Fi networks, management frames handle essential network functions such as device association, authentication, and network control. Action frames represent a specialized subtype of management frames designed to carry vendor-specific or protocol-specific control information between wireless devices.

ESP-NOW specifically utilizes vendor-specific action frames (Type 0, Subtype 0x0D) to transmit user data efficiently while avoiding the overhead associated with standard data frame transmission.

When the `esp_now_send()` function is invoked in the application software, the ESP32 firmware automatically encapsulates the user-defined payload (limited to 250 bytes maximum) within a properly formatted action frame and transmits it to the specified target MAC address. The receiving device's firmware automatically decapsulates the payload and invokes the user-defined callback function `esp_now_recv_cb()` to process the received data.

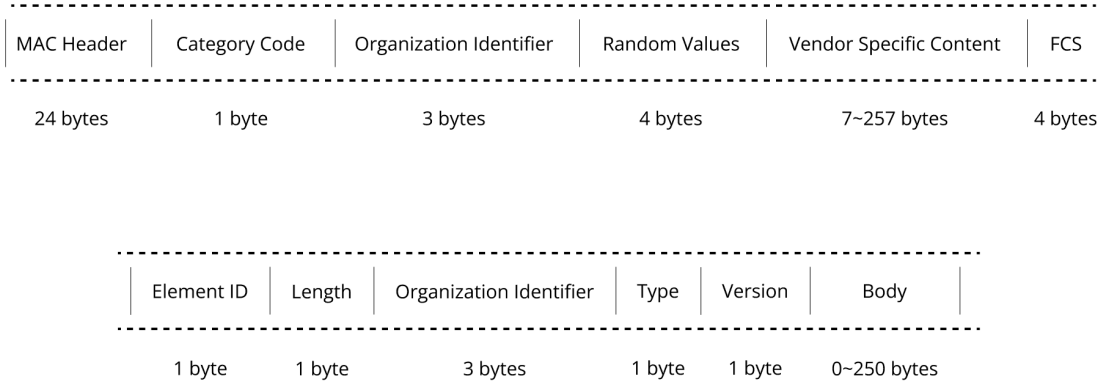


Figure 2.11: Vendor-specific action frame structure utilized by ESP-NOW protocol

Custom V2X Data Packet Structure

For this project implementation, we designed a custom payload format that captures the essential elements of an SAE J2735 Basic Safety Message (BSM) while maintaining a lightweight structure suitable for ESP-NOW's payload limitations. The implemented data structure includes the critical fields necessary for V2V and V2G communication as summarized in Table 2.3.

Data Element	Size (Bytes)	Description
Message ID	1-2	Unique identifier for message type classification (e.g., BSM = 0x01)
Vehicle ID	4-6	Unique identifier for the transmitting vehicle (derived from MAC address)
Timestamp	4	Millisecond counter for synchronization and latency measurement
Latitude	4 (float)	Vehicle's current geographical latitude coordinate
Longitude	4 (float)	Vehicle's current geographical longitude coordinate
Speed	2 (float)	Current vehicle speed in km/h
Heading	2 (float)	Direction of travel in degrees relative to true north
Status Flags	1	Bit flags indicating vehicle state (brake active, turn signals, etc.)

Table 2.3: Custom V2V data packet structure implemented over ESP-NOW protocol

While this custom format does not achieve full compliance with the comprehensive SAE J2735 BSM standard, it successfully captures the essential data elements required to demonstrate core V2V use cases including forward collision warning systems, cooperative awareness applications, and basic traffic management functions.

Security Implementation and Limitations

ESP-NOW incorporates AES-128 encryption capabilities for securing transmitted payloads between communicating peers. The security model requires that all communication participants be pre-configured with an identical 16-byte Local Master Key (LMK) for successful message encryption and decryption.

The encryption process operates as follows: the transmitting device encrypts the payload using the pre-shared LMK before transmission, the encrypted frame is transmitted over the wireless medium, and the receiving device attempts to decrypt the payload using its stored LMK. If decryption fails due to key mismatch or corruption, the received packet is silently discarded without notification.

However, this security approach has significant limitations for production deployments. The static key management system means that the LMK is shared among all network peers, creating a security vulnerability where compromising any single device potentially compromises the entire communication network. Additionally, the basic authentication mechanism provides message confidentiality and integrity but lacks the robust device-level authentication capabilities found in certificate-based security schemes typically required for production V2X systems.

Production V2X deployments typically rely on sophisticated Public Key Infrastructure (PKI) systems with rotating cryptographic keys and digital certificates to ensure secure communication. For this proof-of-concept demonstration, the static LMK mechanism provided sufficient security, but future production deployments should adopt certificate-based authentication and dynamic key management protocols as specified in established V2X security standards.

Project Application ESP-NOW was selected as the primary communication protocol for broadcasting the Twizy’s critical telemetry data including vehicle speed, heading direction, battery state of charge, and GPS position coordinates to nearby vehicles equipped with similar OBU installations and to charging stations equipped with RSU capabilities. The protocol’s inherent simplicity, consistently low latency performance, and independence from external network infrastructure made it particularly well-suited for demonstrating the fundamental principles of cooperative V2V and V2G communication in our experimental setup.

2.4.2 Wi-Fi TCP/IP for V2N Communication

For wide-area Vehicle-to-Network (V2N) connectivity and cloud-based telemetry services, our system employs standard TCP/IP communication over Wi-Fi networks supplemented by cellular-backed internet connectivity. The NVIDIA Jetson Nano connects to a TP-Link M7200 mobile 4G LTE Wi-Fi hotspot, providing reliable internet access for publishing vehicle telemetry data to a centralized ThingsBoard server using the MQTT (Message Queuing Telemetry Transport) protocol. Simultaneously, the charging station’s Raspberry Pi connects to the facility’s Wi-Fi infrastructure and publishes station telemetry data via the same MQTT protocol to enable coordinated V2G monitoring and control.

Network Architecture and Connectivity Features

The implemented network architecture provides several key capabilities for comprehensive V2X communication. Wide-area connectivity is achieved through the Jetson Nano’s connection via the TP-Link M7200 device, which provides cellular (4G LTE) internet access for remote telemetry transmission when fixed network infrastructure is unavailable or unreliable. This ensures continuous data connectivity even during mobile testing scenarios in areas without established Wi-Fi coverage.

MQTT-based telemetry transmission utilizes a lightweight publish/subscribe messaging protocol that is specifically well-suited for Internet of Things (IoT) applications, telemetry data transmission, and intermittent connectivity scenarios. The protocol’s efficiency and reliability make it ideal for publishing both vehicle and charging station operational data to the centralized ThingsBoard monitoring platform.

The system maintains full compatibility with standard Linux networking stacks running on both the Jetson Nano and Raspberry Pi platforms. This compatibility enables the use of well-established MQTT client libraries and standard networking protocols, simplifying development and integration processes while ensuring reliability and maintainability.

The integration approach utilizes MQTT topic-based messaging and JSON payload formats to simplify data ingestion, storage, and visualization within the ThingsBoard platform. This approach

enables comprehensive dashboard creation, historical telemetry data storage, and rule-based processing for automated system responses and alerts.

Implementation in Project Context In our experimental implementation, the Jetson Nano continuously publishes comprehensive vehicle telemetry data including real-time speed measurements, battery state of charge, GPS coordinate information, and diagnostic status data to the ThingsBoard server infrastructure via MQTT protocol over the TP-Link M7200's 4G cellular connection. Concurrently, the charging station's Raspberry Pi, connected to the local facility Wi-Fi network, publishes charging station operational telemetry including power delivery measurements, charging status information, and system health data via MQTT to the same centralized platform.

The ThingsBoard platform serves as the comprehensive telemetry backend, data visualization system, and dashboard platform for the proof-of-concept implementation, enabling real-time monitoring, historical data analysis, remote diagnostics capabilities, and system status visualization for both vehicle and infrastructure components.

For production deployment scenarios, it is strongly recommended to implement MQTT communication over TLS (Transport Layer Security) encryption and establish robust authentication mechanisms using either client certificates or secure token-based credential systems to protect telemetry data transmission and prevent unauthorized system access.

2.5 Vehicle Data Acquisition System

To enable comprehensive V2V and V2G communication capabilities, our system requires real-time extraction of critical vehicle parameters from the Renault Twizy's internal systems. This data acquisition process involves interfacing with the vehicle's Controller Area Network (CAN bus) through the standardized On-Board Diagnostics (OBD-II) connector and implementing sophisticated CAN frame decoding algorithms to extract meaningful operational data.

2.5.1 Controller Area Network (CAN Bus) Fundamentals

The Controller Area Network (CAN bus) represents a robust, standardized communication system that enables multiple Electronic Control Units (ECUs) throughout a vehicle to exchange critical information without requiring centralized computer coordination. This distributed communication architecture allows, for example, the engine control unit to share real-time speed and torque data with the braking system, enabling advanced safety functions such as Anti-lock Braking System (ABS) and traction control operations.

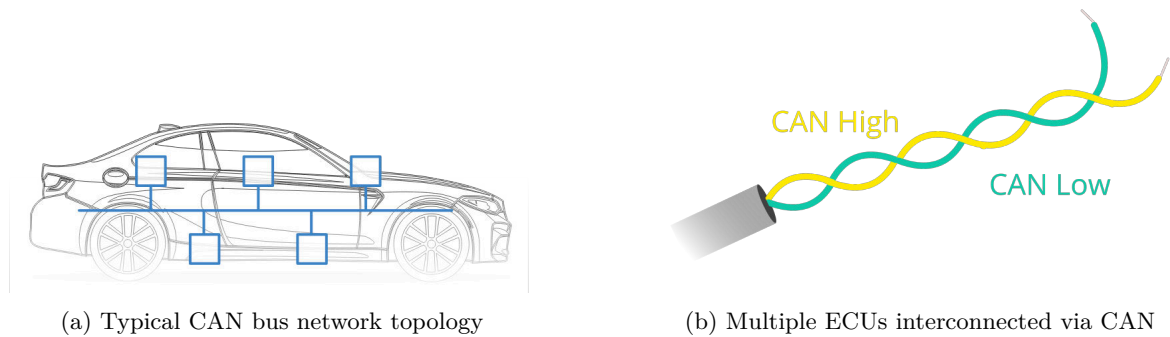


Figure 2.12: CAN bus topology and ECU communication architecture

The physical CAN bus consists of two twisted-pair wires designated as CAN High (CAN_H) and CAN Low (CAN_L), which carry differential signals specifically designed to minimize electromagnetic interference and ensure reliable communication in the electrically noisy automotive environment. The network requires proper termination at both ends using precision 120 Ω resistors to prevent signal reflections and communication errors that could compromise system reliability, as illustrated in Figure 2.13.

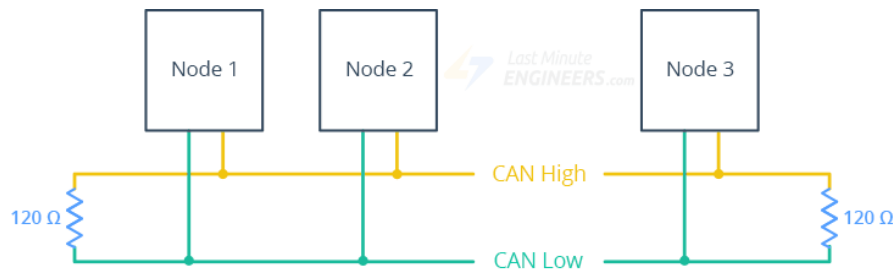


Figure 2.13: Standard CAN bus termination using 120Ω resistors at network endpoints

2.5.2 Electronic Control Unit (ECU) Architecture

Modern vehicles typically incorporate 50 or more specialized ECUs, each responsible for specific operational functions such as engine management, transmission control, battery monitoring, or safety system coordination. Understanding ECU architecture is essential for effective CAN bus data interpretation and system integration.

Each ECU incorporates several key components that enable its participation in the vehicle's communication network. The microcontroller unit (MCU) executes control algorithms and determines what data should be transmitted to other network participants. For example, a temperature monitoring ECU might transmit engine oil temperature data at 5Hz intervals to enable other systems to adjust their operational parameters accordingly.

The CAN controller implements the complex CAN protocol logic including message arbitration, error detection mechanisms, and proper message formatting, effectively offloading this computational complexity from the main MCU and ensuring reliable communication. The CAN transceiver performs the critical function of converting the CAN controller's digital data into the differential electrical signals required for bus transmission while providing electrical protection and isolation between the ECU's internal electronics and the shared bus infrastructure.

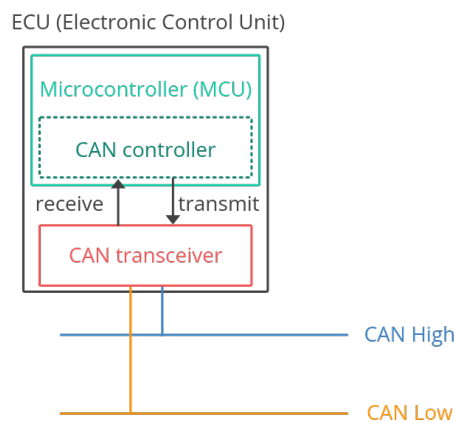


Figure 2.14: Simplified ECU architecture showing MCU, CAN controller, and transceiver integration

2.5.3 CAN Frame Structure and Data Format

Communication over the CAN bus occurs through structured data packets called frames. The standard CAN 2.0A frame format utilizes an 11-bit identifier and provides the foundation for most automotive communication applications.

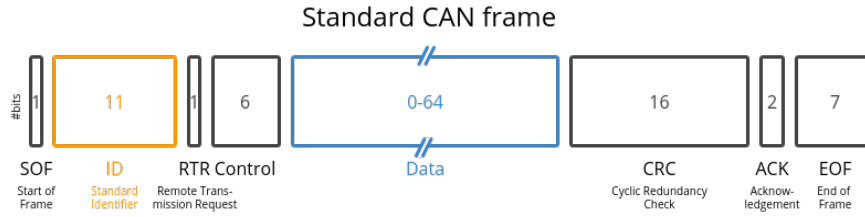


Figure 2.15: Standard CAN 2.0A data frame format and field definitions

The CAN frame structure includes several critical fields that enable reliable communication and data integrity. The Start of Frame (SOF) bit indicates the beginning of a new frame transmission and synchronizes all network participants. The 11-bit Identifier (ID) field serves dual purposes: it identifies the type of data contained in the frame and determines transmission priority, with lower ID values receiving higher priority during bus arbitration.

The Remote Transmission Request (RTR) bit is used to request specific data from another network node, while the Control field contains the Identifier Extension Bit (IDE) and the Data Length Code (DLC) that specifies the number of data bytes in the current frame. The Data field contains the actual payload, supporting 0-8 bytes of information that can be decoded into meaningful vehicle parameters.

Error detection is provided by the Cyclic Redundancy Check (CRC) field, which enables receiving nodes to verify data integrity. The Acknowledgment (ACK) bit confirms that at least one node successfully received the transmitted message, while the End of Frame (EOF) field marks the completion of the current frame transmission.

For our V2X data acquisition purposes, the Identifier and Data fields represent the most critical components, as they enable us to identify and extract specific vehicle signals such as battery state of charge, current consumption, and vehicle speed measurements.

2.5.4 Vehicle Access Through OBD-II Interface

In passenger vehicles, standardized access to the CAN bus is provided through the On-Board Diagnostics II (OBD-II) connector, which follows the SAE J1962 specification. This connector is typically located under the dashboard and provides access to CAN_H (pin 6) and CAN_L (pin 14) signals, along with 12V battery power (pin 16) and electrical ground connections.

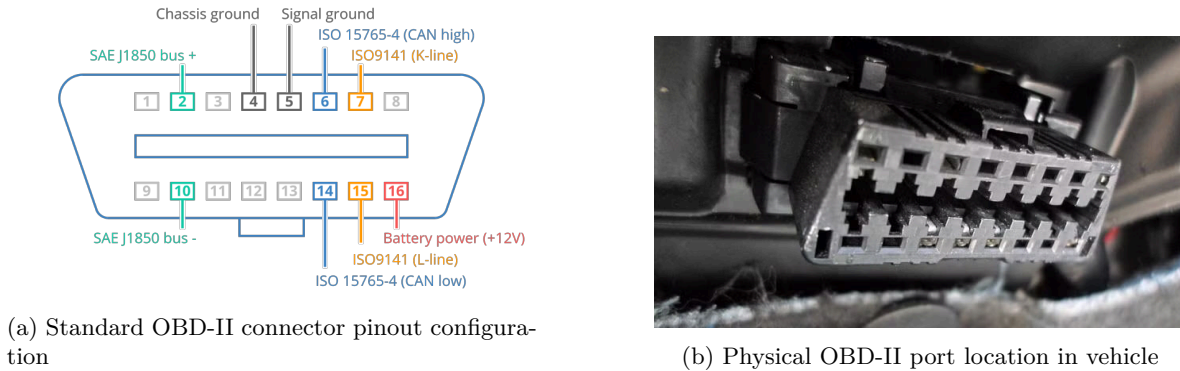


Figure 2.16: OBD-II connector interface for vehicle CAN bus access

2.5.5 Data Acquisition System Implementation

To capture real-time CAN data from the Twizy's internal systems, we implemented a comprehensive data acquisition system using an MCP2515 CAN controller paired with a TJA1050 transceiver module connected directly to the vehicle's OBD-II port. This interface module communicates with an ESP32 microcontroller running custom firmware designed specifically for automotive data acquisition applications.

The ESP32 firmware performs several critical functions including real-time CAN frame reception and timestamping for accurate data correlation, intelligent filtering and storage of relevant messages to reduce processing overhead, and reliable data forwarding to our Jetson Nano computing platform for advanced processing, logging, and analysis operations.

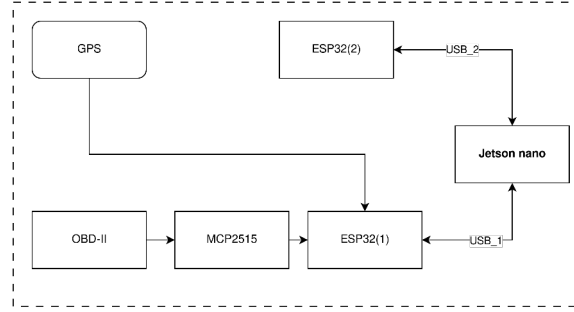


Figure 2.17: Complete CAN data acquisition system architecture using ESP32 and MCP2515 integration

This comprehensive data acquisition setup enables real-time monitoring and logging of essential vehicle parameters including current vehicle speed, battery State of Charge (SOC), system voltage levels, and current consumption patterns. These parameters represent the fundamental data elements required for implementing and validating our V2V and V2G communication experiments. The detailed analysis of captured CAN frame identifiers, data decoding procedures, and signal interpretation methodologies will be presented comprehensively in Chapter 3 of this document.

2.6 OBU and RSU System Architecture

In any comprehensive V2X communication system, the On-Board Unit (OBU) and Road-Side Unit (RSU) represent the two primary system components that enable seamless communication between vehicles and infrastructure. The OBU is permanently installed within the vehicle and assumes responsibility for collecting vehicle operational data, processing this information locally, and establishing communication links with external entities including other vehicles, RSUs, and cloud-based server systems. The RSU is deployed at fixed infrastructure locations such as charging stations, traffic intersections, or highway service areas, serving as a critical gateway that facilitates V2I (Vehicle-to-Infrastructure) communication and data exchange.

2.6.1 Custom OBU Design and Implementation

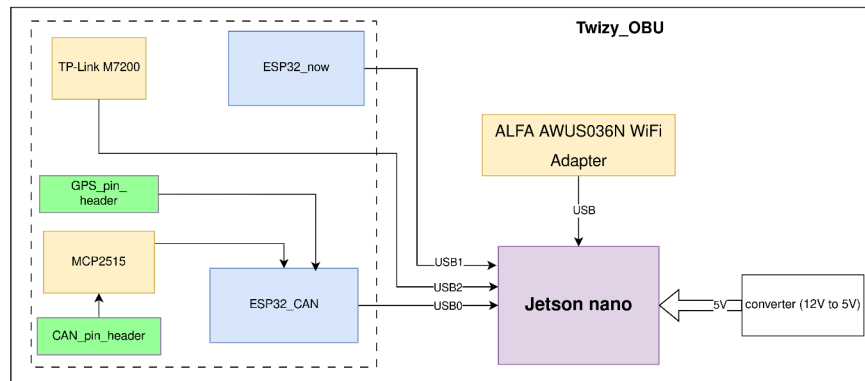


Figure 2.18: Complete architecture of the custom OBU implemented in the Twizy vehicle

Our custom OBU design was specifically developed to collect real-time CAN bus data from the Twizy's internal systems, process this information locally using advanced algorithms, and facilitate reliable message exchange with other OBUs and RSUs throughout the communication network. The hardware architecture incorporates several key components working in close coordination.

The ESP32 microcontroller serves as the primary interface with the MCP2515 CAN controller module, enabling reception of raw CAN frames directly from the vehicle's OBD-II diagnostic port. This microcontroller handles real-time data acquisition, preliminary filtering, and initial processing of vehicle telemetry information.

The NVIDIA Jetson Nano functions as the central processing unit within the OBU architecture, assuming responsibility for receiving processed data from the ESP32 via UART communication, decoding complex CAN messages into meaningful vehicle parameters including speed, battery state of charge, voltage levels, and current consumption, executing sophisticated V2V and V2G communication applications and protocols, and managing data transmission to centralized server systems or local GUI dashboard applications when required for monitoring or analysis purposes.

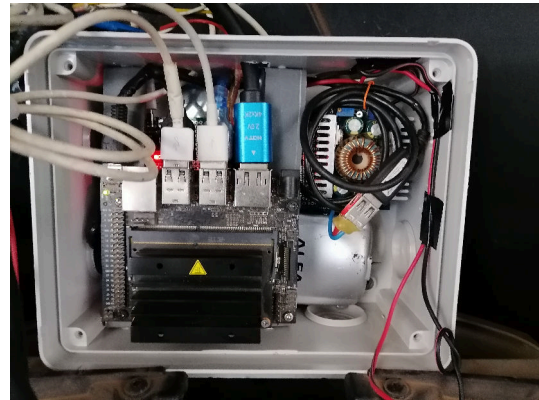
Wireless connectivity capabilities are provided through multiple pathways: ESP32 modules are configured to utilize ESP-NOW protocol for direct V2V communication with other similarly equipped vehicles, while Wi-Fi and Ethernet interfaces enable data transmission to remote server infrastructure for cloud-based monitoring and analysis.

The power supply system is designed to operate reliably from the vehicle's 12V electrical system, incorporating appropriate voltage regulation and power conditioning circuitry to provide stable power to both the Jetson Nano and ESP32 components while ensuring electrical safety and system reliability. Figure 2.18 illustrates the comprehensive architecture of our custom OBU design, showing the integration of all major components and their interconnections.

2.6.2 OBU Physical Prototype Development



(a) Assembled OBU prototype prepared for vehicle installation



(b) Internal component layout of the OBU prototype

Figure 2.19: Physical prototype of the custom OBU system installed in the Twizy vehicle

We successfully developed and constructed a physical prototype of the OBU system and integrated it into the Twizy test vehicle. The prototype incorporates a NVIDIA Jetson Nano computing module, an ESP32 microcontroller, a complete CAN interface consisting of MCP2515 controller and TJA1050 transceiver, and comprehensive power regulation circuitry, all housed within a protective enclosure designed to withstand the automotive operating environment.

2.6.3 EEBLAB Custom EVCS RSU Design and Architecture

The Road-Side Unit (RSU) implemented in our system serves as a critical communication bridge between vehicles and charging infrastructure, facilitating seamless data exchange between the vehicle's OBU and backend server systems. The RSU's primary operational functions include receiving data packets from vehicle OBUs using ESP-NOW or Wi-Fi communication protocols, aggregating and processing received data before forwarding it to centralized server systems or cloud platforms for advanced analysis and storage, and relaying control commands such as charging initiation, termination, or parameter adjustment from the infrastructure back to connected vehicle OBUs.



Figure 2.20: EEBLAB custom Electric Vehicle Charging Station with integrated RSU

Our RSU prototype implementation was intentionally designed with a simplified architecture compared to the vehicle-mounted OBU, focusing on essential functionality while maintaining cost-effectiveness and reliability. The system architecture consists of three primary components working in coordination to provide comprehensive charging station monitoring and communication capabilities.

The PZEM-004T power measurement sensor serves as the primary data acquisition component, continuously measuring critical electrical parameters including output power delivery, current flow, voltage levels, and total energy transferred to connected vehicles. This sensor provides the fundamental data required for V2G power management and grid integration analysis.

The ESP32 microcontroller interfaces directly with the PZEM-004T sensor via UART communication, collecting measured electrical data and implementing V2G communication protocols. The ESP32 processes the acquired information and forwards it to the Raspberry Pi for additional processing and network transmission.

The Raspberry Pi provides a comprehensive Linux-based computing environment that handles advanced message processing, data logging, and network communication functions. The Raspberry Pi publishes collected power measurement data and charging station status information to the EE-BLAB ThingsBoard server infrastructure using the MQTT protocol, enabling real-time monitoring and historical data analysis.

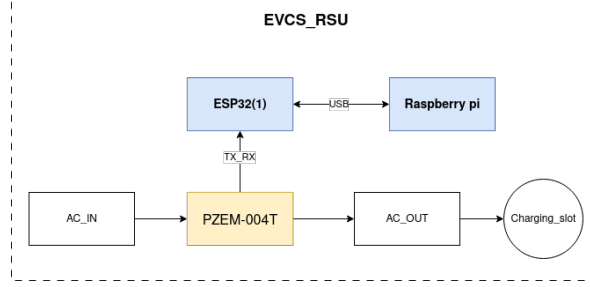


Figure 2.21: Detailed architecture of the EVCS RSU prototype system

Figure 2.21 presents the complete system architecture of the RSU prototype, illustrating the integration of power measurement, data processing, and communication components that enable comprehensive V2G functionality.

2.7 System Integration and Communication Flow

The complete V2X communication system integrates vehicle-mounted OBUs with infrastructure-based RSUs to create a comprehensive connected vehicle ecosystem. The OBU continuously monitors vehicle operational parameters and transmits this information using multiple communication protocols depending on the target recipient and application requirements.

For V2V communication scenarios, the OBU broadcasts vehicle telemetry data including position, speed, heading, and battery status using ESP-NOW protocol to nearby vehicles equipped with compatible OBUs. This enables implementation of cooperative awareness applications, collision avoidance systems, and traffic efficiency optimization algorithms.

For V2G communication, the OBU establishes direct communication with charging station RSUs to coordinate power transfer operations, monitor charging progress, and implement grid integration functions. The RSU aggregates data from multiple vehicles and forwards comprehensive charging station status information to centralized monitoring systems.

V2N communication enables the OBU to maintain connectivity with cloud-based services for remote monitoring, software updates, and integration with broader transportation management systems. This connectivity is maintained through the TP-Link M7200 mobile hotspot, ensuring reliable communication even in areas without fixed network infrastructure.

2.7.1 Summary and System Benefits

The integration of custom OBU and RSU components enables comprehensive real-time data exchange between the Twizy test vehicle and external infrastructure systems. The OBU ensures that critical vehicle parameters are continuously collected, processed locally using advanced algorithms, and transmitted reliably to appropriate recipients based on application requirements.

The RSU functions as an intelligent gateway to infrastructure systems, enabling sophisticated V2I and V2G use cases including coordinated charging management, grid integration services, and infrastructure-based traffic management. Together, these components provide a complete foundation for demonstrating and validating advanced V2X communication scenarios including Vehicle-to-Vehicle cooperative applications and Vehicle-to-Grid energy management systems.

The modular design approach ensures that individual components can be upgraded or replaced as technology evolves, while the use of standard communication protocols and interfaces ensures compatibility with future V2X infrastructure deployments and commercial vehicle systems.

Vehicle-to-Vehicle (V2V) Implementation

3.1 Background on V2V

For decades, vehicles have been relatively isolated systems: initially receiving only radio signals for entertainment, and more recently becoming connected to support navigation and infotainment services. Today, advanced communication technologies are becoming central to vehicle operation, enabling enhanced road safety, improved traffic efficiency, and a better driving experience — all of which are stepping stones toward fully autonomous vehicles [17].

The domain of vehicle communications is now collectively referred to as **V2X**, which encompasses several subcategories:

- **V2V** : Direct communication between vehicles to exchange speed, position, and other safety-critical data.
- **V2I** : Communication with roadside infrastructure such as traffic lights and charging stations.
- **V2G** : Interaction between electric vehicles and the power grid for charging and energy management.
- **V2P** : Emerging technology enabling vehicles to communicate with vulnerable road users.

As the industry progresses toward autonomous driving, V2V communication is expected to evolve from simple driver alerts to active vehicle control, such as automatic braking or evasive steering. Most V2V systems are envisioned as forming a **mesh network**, using technologies such as Dedicated Short-Range Communications (DSRC). DSRC operates in the 5.9 GHz band and typically offers a range of about 300 meters (approximately 10 seconds of warning time at highway speeds). With multi-hop communication, this range can extend to around 1 mile, allowing early detection of traffic events in congested areas.

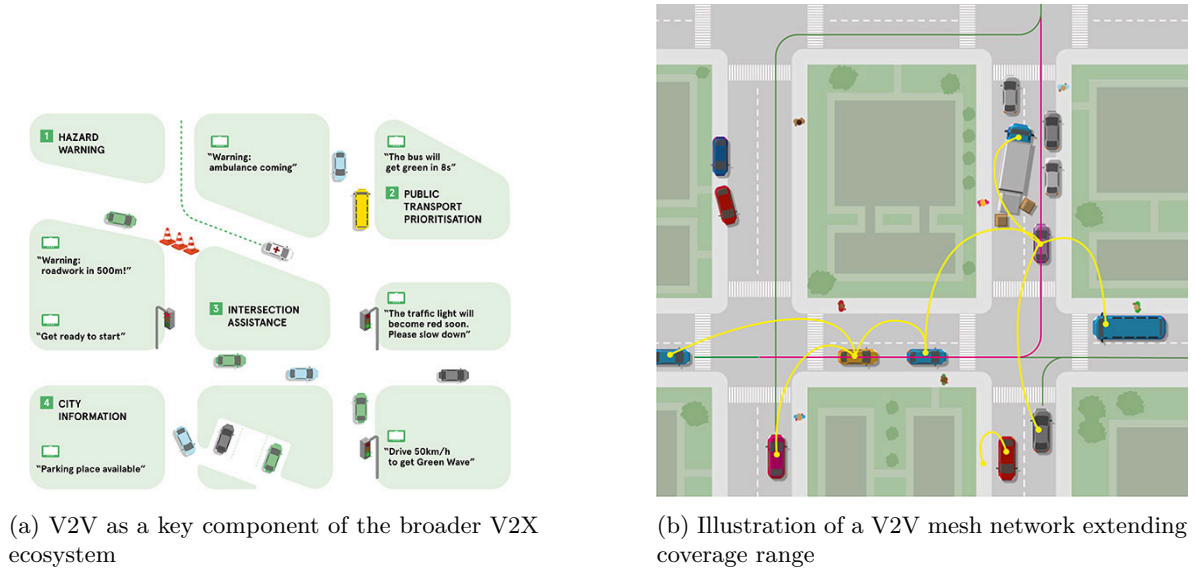


Figure 3.1: Overview of V2V technology and its networking capabilities

3.2 Proof of Concept Implementation

In this section, we present our **low-cost V2V communication proof of concept**, implemented on a Renault Twizy electric vehicle. The implementation focuses on:

- The hardware and software architecture of the On-Board Unit (OBU).
- The workflow of data exchange between multiple OBUs.
- Experimental results and the challenges encountered during development.

3.3 Implementation on Twizy OBU

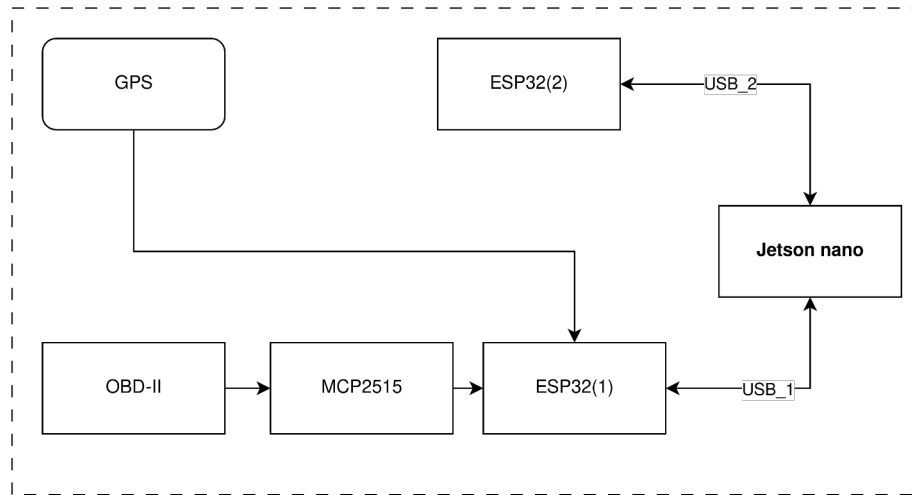


Figure 3.2: Block diagram of the CAN data acquisition setup using ESP32 and MCP2515

The On-Board Unit (OBU) 2.19a developed for the Renault Twizy served as the central platform for data collection and V2V communication. The implementation process can be divided into three main parts: **data acquisition**, **data processing** and **wireless communication**.

Data Acquisition

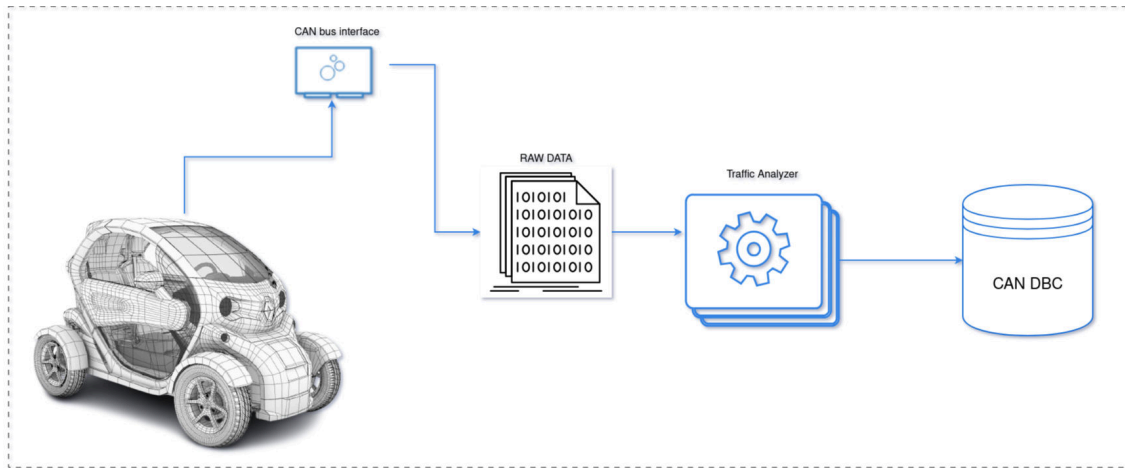


Figure 3.3: CAN Database (DBC) creation flow

This was one of the most critical steps of the project, as no official documentation from the manufacturer (Renault) is publicly available listing all Service Identifiers (SIDs) or CAN message details for the Twizy.

Vehicle manufacturers consider their CAN databases — including SIDs, PIDs, message identifiers, and decoding formulas — as proprietary intellectual property. Publishing this data could:

- Allow competitors to reverse-engineer their systems.
- Expose security vulnerabilities by revealing how to query or control vehicle functions.
- Undermine their diagnostic tool business model by enabling unauthorized third-party solutions.

To overcome this challenge, we leveraged work from the open-source community [23]. Several hobbyists [26] have successfully reverse-engineered the Twizy CAN bus by connecting to the OBD-II port using CAN interfaces (PCAN, Arduino with MCP2515, USB2CAN, etc.) and capturing traffic with tools such as Wireshark, CANalyzer, or SocketCAN. Their findings have been compiled into an open-source **Twizy CAN Database (DBC)** (Figure 3.3).

The Twizy CAN DBC 3.4 essentially contains:

- **CAN IDs (e.g., 0x0080, 0x0081, 0x0155):** Identifiers for each message, encoding source ECU, message type, and priority.
Example:
0x0155 → Core battery status (SOC, current, charging power)
0x0196/0x019F → Motor power, speed, temperature
0x059B → Drive status (gear, brake, accelerator)
- **Transmission Interval (ms):** The period at which each message is broadcast.
- **Mode:** Indicates data flow type — **D** = broadcast (no response expected), **D+C** = data + command (bidirectional).
- **Source Module:** ECU that transmits the message (e.g., BMS, CON, CHG, DSP).
- **Message Name:** Human-readable description.
- **Data Length:** Number of data bytes.
- **Byte Position:** The byte offset where the parameter is located.
- **Parameter Description:** Meaning of the data.

- **Decoding Formula:** Method to convert raw data to human-readable values.

0155	10	D+C	BMS	BMS Status 1	8	1	Charging Power Control (BMS - CHG)	Steps 7..0: Charging power = Step * 300W or Charging current = Step * 5A (at battery), Step 0 at 100% SOC
0155	10	D+C	BMS	BMS Status 1	8	2+3	Traction Battery Current	Lower 12 bits: TDOh = 2000d zero point, Deviation * 16 = approx. Power [W], negative=discharge, Current = Deviation * 4 [A], Upper 4 bits: Charger handshake
0155	10	D+C	BMS	BMS Status 1	8	4	Phase Status	94h = Init/Exit phase = Invalid data // 54h = Valid data
0155	10	D+C	BMS	BMS Status 1	8	5+6	State of Charge (SOC)	SOC% = Decimal value * 400, Only upper 13 bits used; End of charge = 9C50h
0155	10	D+C	BMS	BMS Status 1	8	7	Reserved	Fixed 07
0155	10	D+C	BMS	BMS Status 1	8	8	Unknown Parameter	Range 5Fh - 73h
0196	10	D	CON	Energy Status 2	8	4+5	Power Request / Torque / Slip	8 bit B4 + 1 bit B5: Rest value 254, Current power request [W] = Deviation * 250, negative = braking/reverse
0196	10	D	CON	Energy Status 2	8	6	Motor Temperature	Temperature [°C] = Value - 40
019F	10	D	CON	Energy Status 3	8	1	Reserved	Fixed FFh?
019F	10	D	CON	Energy Status 3	8	2	Reserved	Fixed FFh?
019F	10	D	CON	Energy Status 3	8	3+4	Speed (+/-)	8 bit B3 + 4 bit B4: Rest value 7DOh=2000, RPM=(Value-2000)*10, kph=RPM*7250*60 (T80)
019F	10	D	CON	Energy Status 3	8	5	Reserved	Fixed 38h?
019F	10	D	CON	Energy Status 3	8	6	Reserved	Fixed FFh?
019F	10	D	CON	Energy Status 3	8	7	Reserved	Fixed 40h?
019F	10	D	CON	Energy Status 3	8	8	Reserved	Fixed FEh?
0423	100	D+C	CHG	Charger Status	8	1	Charger On/Off Status	03h = ON, 00h = OFF
0423	100	D+C	CHG	Charger Status	8	2	Unknown Parameter	?
0423	100	D+C	CHG	Charger Status	8	7+8	Unknown Counter	Rises to 00E0 after power-on and remains relatively constant
0424	100	D+C	BMS	BMS Status 2	8	1	Charging Protocol (BMS - CHG)	1xh = see charging protocol sheet / 9xh = Battery error

Figure 3.4: Open-source Twizy CAN DBC

Having access to this database allowed us to know exactly what data to expect on the CAN bus. The OBU continuously captures CAN frames (Figure 3.5a) using the MCP2515 and ESP32(1) modules (see Figure 3.2) and forwards them to the Jetson Nano. The data is then logged into two CSV files for further processing and analysis.

```

080 0
155 8 07 97 CC 54 62 D8 00 6E
19F 8 FF FF 7D 0F 38 FF 40 FE
19F 8 FF FF 7D 0F 38 FF 40 FE
19F 8 FF FF 7D 0F 38 FF 40 FE
19F 8 FF FF 7D 0F 38 FF 40 FE
080 0
599 8 00 00 0D AD FF 19 00 00
155 8 07 97 CC 54 62 D8 00 6E
599 8 00 00 0D AD FF 19 00 00
155 8 07 97 CB 54 62 D8 00 6E
599 8 00 00 0D AD FF 19 00 00
155 8 07 97 CB 54 62 D8 00 6E
436 6 00 00 00 38 00 00
598 8 20 00 64 00 40 14 00 90
436 6 00 00 00 38 00 00
598 8 20 00 64 00 40 14 00 90

```

(a) CAN frames captured with the MCP2515 and ESP32(1)

```

080 0
155 8 07 97 CC 54 62 D8 00 6E
19F 8 FF FF 7D 0F 38 FF 40 FE
19F 8 FF FF 7D 0F 38 FF 40 FE
19F 8 FF FF 7D 0F 38 FF 40 FE
19F 8 FF FF 7D 0F 38 FF 40 FE
080 0
599 8 00 00 0D AD FF 19 00 00
155 8 07 97 CC 54 62 D8 00 6E
599 8 00 00 0D AD FF 19 00 00
155 8 07 97 CB 54 62 D8 00 6E
599 8 00 00 0D AD FF 19 00 00
155 8 07 97 CB 54 62 D8 00 6E
436 6 00 00 00 38 00 00
598 8 20 00 64 00 40 14 00 90
436 6 00 00 00 38 00 00
598 8 20 00 64 00 40 14 00 90

```

(b) Structure of CAN frames captured with the MCP2515 and ESP32(1)

Figure 3.5: Overview of CAN frames captured and their structure

Data Processing and Wireless Communication

After acquisition, the CAN frames are parsed and processed to make them suitable for storage, analysis, and wireless transmission. The ESP32 extracts relevant signals from each CAN frame and formats them into compact data packets.

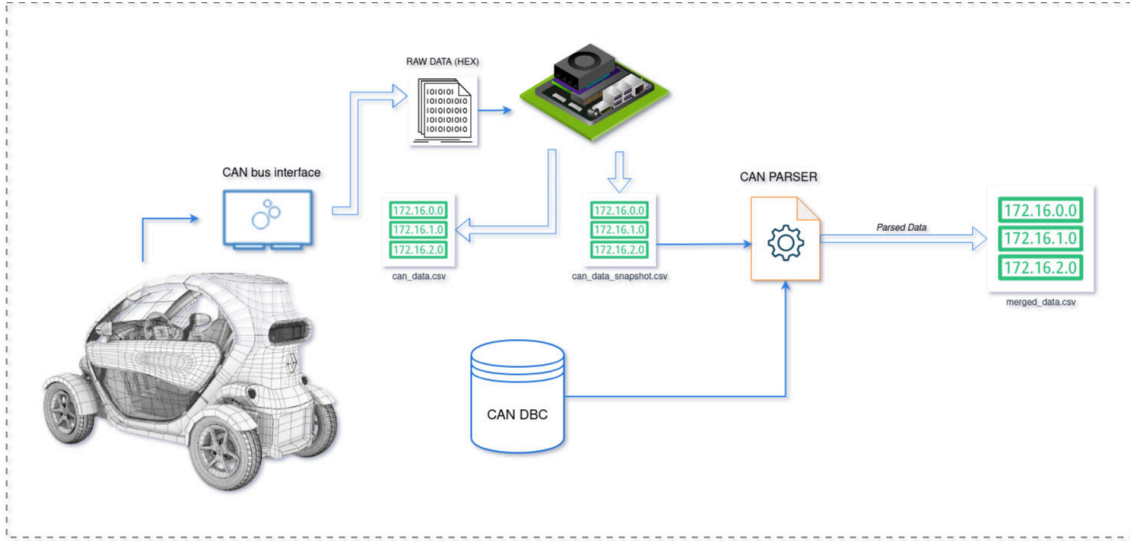


Figure 3.6: CAN bus data parsing flow

Once received by the Jetson Nano, the data is logged into two CSV files (Figure 3.7):

- **can_data.csv** A continuously updated log of all incoming CAN messages.
- **can_data_snapshot.csv** A snapshot updated every 4 seconds, used by other programs for data processing without interfering with the main logging stream.

A dedicated parser program (ESPnow_EVCS_V2G.c) running on the Jetson reads the snapshot file, decodes the hexadecimal CAN frames using a library built from the Twizy CAN DBC, and converts them into human-readable values. These decoded values are then stored in a third file, **merged_data.csv**, as shown in Figure 3.8.

timestamp	can_id	dlc	data_hex	gps_lat	gps_lon	gps_alt	gps_valid
1756217847.56268	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.57391	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.58493	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.5964	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.60715	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.61799	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.62942	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.63983	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.65094	423	8	03 2B FF FF 00 E0 00 E2	33.986301	-6.725	149.5	15
1756217847.66199	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.67398	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.68466	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.69464	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.70577	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.71666	425	8	0A 37 44 FF FE 66 01 33	33.986301	-6.725	149.5	15
1756217847.72767	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.73867	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.74986	423	8	03 2B FF FF 00 E0 00 E2	33.986301	-6.725	149.5	15
1756217847.76066	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.77184	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.78286	507	7	FF FF 00 58 10 30 00	33.986301	-6.725	149.5	15
1756217847.79402	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15
1756217847.80477	155	8	07 97 CD 54 90 98 00 72	33.986301	-6.725	149.5	15

Figure 3.7: CAN and GPS data captured and stored on the Jetson Nano

timestamp	soc	current	gear	motor_active	accelerator	brake	cap_voltage	motor_speed	odometer	range	battery_voltage
1753441690.88019	81	-2.5	Neutral	1	0	0	56.5	0	3506	29	57.5
1753441696.89989	81	-2.8	Reverse	1	0	1	56.5	0	3506	29	57.5
1753441702.91545	81	-2.5	Reverse	1	0	1	56.5	0	3506	29	57.5
1753441708.93307	81	-2.8	Reverse	1	0	1	56.5	0	3506	29	57.5
1753441714.84921	81	-2	Drive	1	0	0	56.5	0	3506	29	57.5
1753441720.91857	81	-2.5	Reverse	1	16	0	56.5	-1.8	3506	29	57.5
1753441726.98774	81	-1.8	Reverse	1	9	0	56.5	-3.6	3506	29	57.5
1753441732.75021	81	-25	Reverse	1	45	0	56.5	-3.6	3506	29	57.5
1753441739.02315	81	-13.8	Drive	1	86	0	55.5	5.3	3506	29	57.5
1753441744.9904	81	-16.8	Drive	1	87	0	56	8.8	3506	29	57.5
1753441751.0597	81	-43.2	Drive	1	0	0	56.5	8.8	3506	29	57.5
1753441757.07728	81	-19	Drive	1	44	0	56.5	14.1	3506	29	57.5
1753441763.09542	81	-37.8	Drive	1	51	0	56	14.1	3506	29	57.5
1753441768.80771	81	-2.5	Drive	1	51	0	56	15.8	3506.1	29	57.5
1753441775.13248	81	-25.8	Drive	1	0	1	56.5	3.5	3506.1	29	57.5
1753441781.09791	80	5	Drive	1	10	0	56.5	14.1	3506.1	29	57.5
1753441787.16625	80	-14.8	Drive	1	41	0	56	15.8	3506.2	29	57.5
1753441793.18482	80	15	Drive	1	41	0	56	12.3	3506.2	29	57.5
1753441798.89719	80	-2.5	Drive	1	35	0	56.5	10.5	3506.2	29	57.5
1753441805.22047	80	14	Drive	1	21	0	56.5	10.5	3506.2	29	57.5
1753441811.18821	80	-45	Drive	1	0	1	56.5	5.3	3506.2	29	57.5
1753441817.25723	80	-32.2	Drive	1	52	0	56	10.5	3506.3	29	57.5
1753441823.27526	80	38.5	Drive	1	52	0	56	26.4	3506.3	29	57.5
1753441828.93629	80	-47	Drive	1	52	0	56	5.3	3506.3	29	57.5
1753441835.31163	80	-143.2	Drive	1	74	0	55	26.4	3506.3	29	57.5

Figure 3.8: Example content of parsed and merged CAN data

In addition to local logging, the parser program forwards the decoded data to **ESP32(2)** via USB serial. ESP32(2) is responsible for transmitting this data to other OBUs or to the Roadside Unit (RSU) using ESP-NOW, enabling low-latency vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication.

For cloud integration, a secondary process (`SendToServer.c`) continuously monitors `merged_data.csv` and publishes selected parameters (vehicle speed, braking status, battery SOC, GPS coordinates, etc.) to EEELAB's local ThingsBoard server using MQTT. This ensures that all nodes in the V2X ecosystem have access to critical information for real-time decision-making — for example, collision avoidance, state-of-charge monitoring, and theft detection.

Software Stack The software running on the ESP32 modules was developed in C using the ESP-IDF framework, providing low-level control and efficient resource usage. On the Jetson Nano, multithreading (via POSIX `pthread`s) was employed to handle concurrent tasks such as data logging, parsing, serial communication, and MQTT publishing. This modular design ensured reliable performance and easier debugging during testing. The overall architecture of the system is summarized in Figure 2.18.

V2N Communication & Dashboard Integration

In addition to local V2V broadcasting, the OBU was configured to support **Vehicle-to-Network (V2N)** communication. This was achieved by forwarding selected decoded CAN parameters (State of Charge, vehicle speed, gear status, GPS position) from the Jetson Nano to a local **ThingsBoard** server using the MQTT protocol. A dedicated service continuously monitored the `merged_data.csv` file and published updates every four seconds to the server. On ThingsBoard, a custom dashboard was created to visualize the data in real time, as shown in Figure 3.10. The dashboard includes widgets for:

- **Battery State of Charge (SoC)** – represented as a gauge.
- **Vehicle Speed and Gear Selection** – displayed as digital indicators.
- **GPS Position** – plotted on a map widget for vehicle tracking.

This V2N integration demonstrates how the developed OBU can act as a bridge between the vehicle and cloud-based applications, enabling remote monitoring, fleet management, and predictive maintenance use cases. It also lays the groundwork for future deployment of cooperative mobility services, where vehicle data can be shared with infrastructure operators or other vehicles through a central platform.

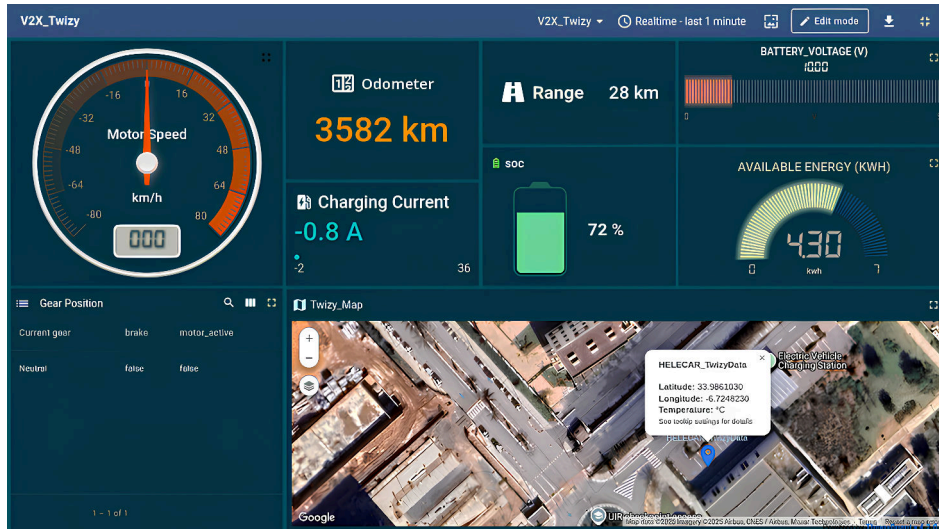


Figure 3.9: Twizy status dashboard built on ThingsBoard

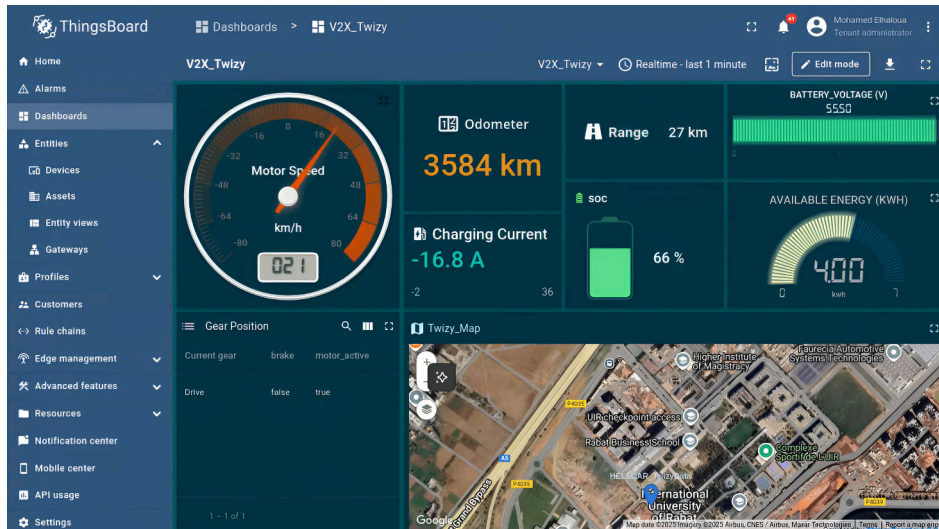


Figure 3.10: Twizy status update dashboard built on ThingsBoard while driving the vehicle across UIR's campus

3.4 Use Cases Tested & Results

To validate the functionality and robustness of our V2V communication system, several test scenarios were designed and executed:

- **ESP-NOW Connectivity Test:** We first implemented a basic communication link between multiple ESP32 devices to ensure that they could broadcast dummy data packets reliably. The test confirmed that the devices could exchange messages with minimal packet loss and negligible latency.
- **Vehicle Data Broadcasting:** After validating basic connectivity, we enabled the Twizy OBU to broadcast real vehicle data — including State of Charge (SoC), vehicle speed, gear selection, braking status, and other relevant parameters — to neighboring OBUs (Figure 3.11). An external mobile ESP32 successfully received these broadcasts and was also able to send back simulated data of the same type (SoC, speed, brake status, etc.), which the Twizy OBU received and processed correctly. This confirmed reliable bidirectional communication between the Twizy

- **Latency:** The average end-to-end packet latency was measured at approximately 10–15 ms, which is adequate for non-critical V2V scenarios.
- **Range:** Reliable communication was achieved up to 80–100 m in open environments. Beyond this range, packet loss increased noticeably.
- **Data Integrity:** Less than 2% packet loss was observed within the effective range, and received packets were verified for consistency.

3.5 Challenges & Limitations

Several limitations and open issues were identified during testing:

- **Limited Use-Case Testing:** Due to the time constraints of the internship, only a subset of potential use cases was tested. Critical scenarios such as multi-vehicle convoy tests, high-speed driving, emergency braking warnings, and accident detection were not evaluated.
- **Lack of Standardization:** The communication format used was custom-designed and not compliant with standardized V2X protocols (e.g., IEEE 802.11p, C-V2X). This may limit interoperability with commercial systems.
- **Security Concerns:** Although ESP-NOW provides optional encryption, no robust authentication or message-signing mechanism was implemented. Since the system used broadcast mode, it remains vulnerable to spoofing or malicious data injection. Future work should include a solid security and data encryption such as using AES-encrypted payloads, message signing, and authentication to mitigate spoofing.
- **Scalability:** The proof of concept was tested with only two OBUs. The system's performance in a dense vehicular network remains unknown, which require in upcoming work stress-testing with simulated traffic using NS-3 or SUMO.

These challenges open several avenues for future work, including standardizing the communication protocol, implementing stronger cryptographic security, and performing large-scale field trials with multiple vehicles.

3.6 Chapter Summary

In this chapter, we presented the implementation and validation of a low-cost V2V communication system on the Renault Twizy. The OBU successfully captured real-time CAN bus data, processed it, and broadcasted key parameters (SoC, speed, gear status, etc.) to neighboring OBUs using ESP-NOW. The results confirmed the feasibility of our approach, demonstrating acceptable latency, range, and reliability for proof-of-concept purposes. Despite its limitations, the system lays a strong foundation for further development of cooperative driving and energy management applications, as well as integration with V2G scenarios explored in the following chapter.

Vehicle-To-Grid (V2G) Implementation

4.1 Background on V2G

Vehicle-to-Grid (V2G) technology represents a transformative approach to energy management that emerged from the convergence of two critical global trends: the electrification of transportation and the modernization of electrical grids. First conceptualized by Willett Kempton at the University of Delaware in the early 2000s, V2G technology addresses the growing need for grid flexibility as renewable energy sources become more prevalent.

The rapid evolution of electric vehicles (EVs) has brought about a paradigm shift in both the transportation and energy sectors. As EV adoption increases, so does the need for innovative solutions to manage energy consumption and ensure grid stability. This is where Vehicle-to-Grid (V2G) technology comes into play. V2G, a key subset of Vehicle-to-Everything (V2X), focuses on the interaction between EVs and the electrical grid.

Vehicle-to-Grid communication enables electric vehicles to interact with the power grid in a bidirectional manner. This interaction allows EVs not only to draw energy from the grid for charging but also to return excess energy stored in their batteries back to the grid. V2G is made possible through advanced hardware, software, and standardized communication protocols that ensure seamless energy exchange and real-time data sharing between EVs and grid operators.

The concept is based on using EVs as mobile energy storage units. By integrating EVs into the grid, V2G helps balance energy supply and demand, reduce peak loads, and enhance overall grid resilience. This is particularly important for the integration of renewable energy sources, where fluctuations in power generation can pose challenges to grid stability.

Key Components of Vehicle-to-Grid Communication

- **Bidirectional Chargers:** These specialized chargers enable energy to flow in both directions—charging the EV battery and discharging energy back to the grid. They include inverters that convert DC energy from the battery into AC energy compatible with the grid.
- **Communication Protocols:** V2G communication relies on standardized protocols such as ISO 15118, Open Charge Point Protocol (OCPP), and IEEE 2030.5. These protocols ensure interoperability between EVs, charging stations, and grid systems, while also supporting secure and efficient data exchange.
- **Energy Management Systems (EMS):** EMS play a crucial role in monitoring and controlling energy flow between EVs and the grid. They analyze real-time data, optimize energy usage, and contribute to maintaining grid stability.
- **Smart Grid Infrastructure:** A robust smart grid infrastructure is essential for V2G implementation. Smart grids are equipped with sensors, meters, and communication networks that enable real-time monitoring and management of energy flows.
- **EV Batteries:** The EV battery is the core component of V2G. High-capacity batteries with advanced storage capabilities are required to enable efficient and reliable energy exchange.

The V2G implementation consists of two primary subsystems: the vehicle-side On-Board Unit (OBU) and the Electric Vehicle Charging Station Road Side Unit (EVCS-RSU) 4.2. The vehicle subsystem comprises a Jetson Nano embedded computer serving as the central processing unit, coupled with an ESP32 microcontroller responsible for wireless communication protocols. This configuration enables real-time processing of vehicle telemetry data while maintaining low-latency communication capabilities essential for V2G operations. The charging station infrastructure integrates a Raspberry Pi single-board computer with an ESP32 wireless communication module and PZEM-004T power measurement devices. This arrangement provides comprehensive monitoring of electrical parameters during charging sessions while enabling bidirectional communication with approaching vehicles. The system supports multiple communication protocols including ESP-NOW for short-range device-to-device communication, WiFi for local network connectivity, and cellular networks (3G/4G/5G) for wide-area communication through cloud-based services. The data acquisition framework encompasses multiple sources critical for V2G functionality:

1. **Vehicle Side:** Jetson Nano + ESP32 (Twizy_OBU - On-Board Unit)
2. **EVCS Side:** Raspberry Pi + ESP32 + PZEM-004T power meter (EVCS_RSU - Electric Vehicle Charging Station Road Side Unit)
3. **Communication Protocols:** ESP-NOW (2.4GHz wireless), WiFi, and Cellular (3G/4G/5G)
4. **Data Sources:** CAN bus telemetry, GPS positioning, and electrical power measurements

4.4 Vehicle-to-Grid Use Cases and Applications

The V2G system developed during this internship addresses multiple application scenarios that demonstrate the versatility and potential impact of bidirectional vehicle-grid communication. The primary implementation focuses on establishing intelligent wireless communication between vehicles and charging infrastructure using various wireless protocols, each optimized for specific operational ranges and data requirements. A significant advancement in this project is the development of an intelligent recommendation system that considers multiple driver preferences including time constraints, cost optimization, comfort factors, and service quality metrics. This system analyzes real-time vehicle state parameters such as current State of Charge (SOC), geographic position, battery State of Health (SOH), and vehicle speed to recommend the most appropriate charging station for each specific journey. Furthermore, the system incorporates predictive algorithms that calculate *optimal driving speeds* required to reach selected charging stations while considering energy consumption patterns and battery depletion rates. An innovative application area explored in this project involves utilizing V2G technology for medical and emergency support applications, particularly in home healthcare scenarios. Electric vehicles can function as mobile energy storage systems, providing critical backup power during grid outages or in areas with unreliable electrical infrastructure. This capability is particularly valuable for powering essential medical equipment including:

- Oxygen concentrators for respiratory support
- Dialysis machines for renal treatment
- Medication refrigeration systems (e.g., insulin storage)
- Ventilators and Continuous Positive Airway Pressure (CPAP) machines



(a) EEBLAB custom EVCS slot2



(b) EEBLAB custom EVCS slots, one for charging the vehicle and the other for charging small devices under 2kW

Figure 4.3: Overview of the EEBLAB custom charging station for vehicles and small devices

The healthcare application of V2G technology addresses three critical operational requirements. First, it provides emergency resilience during natural disasters, power grid failures, or in rural areas with inadequate electrical infrastructure, ensuring continuous operation of life-critical medical devices. Second, it reduces dependence on centralized healthcare facilities by enabling reliable home-based medical care with guaranteed power availability. Third, it facilitates integration with smart home energy management systems that automatically prioritize medical equipment during peak demand periods or power outages, ensuring uninterrupted healthcare delivery.

The charging station implementation as shown on figure 4.1 represents a comprehensive approach to renewable energy integration and intelligent power distribution. The system architecture begins with solar photovoltaic panels that generate direct current (DC) electricity, which is monitored and optimized through Maximum Power Point Tracking (MPPT) controllers before being stored in 24V battery banks. The stored renewable energy is subsequently converted to alternating current (AC) through DC-AC inverters to provide standardized vehicle charging capability. The charging infrastructure incorporates two distinct power sources to ensure operational reliability and charging flexibility figure 4.3b. Slot 2 provides energy derived from the solar panel and battery storage system, representing the renewable energy pathway and demonstrating sustainable charging practices. Slot 1 connects directly to the utility grid (mains power), offering an alternative charging source when solar energy generation is insufficient or during periods of high demand. This dual-source configuration provides drivers with reliable charging options while demonstrating different energy sourcing strategies within the V2G framework.



Figure 4.4: Twizy charging from EEBLAB custom EVCS's slot2

V2G Communication Workflow

The V2G communication protocol operates across two distinct operational ranges (see figure 4.1), each serving specific functional requirements within the overall system architecture. Long-range communication between vehicles and charging stations initiates while vehicles are still traveling on roadways, potentially several kilometers from the charging facility. The On-Board Unit (OBU) in the vehicle and the Road Side Unit (RSU) at the charging station exchange operational information through a centralized server infrastructure, enabling drivers to access real-time data regarding charging station availability, slot occupancy status, power delivery capabilities, precise geographic coordinates, and current pricing structures.

This long-range wireless communication is facilitated through internet connectivity, with both vehicle and charging station systems connected to a centralized server via WiFi or cellular 5G networks, depending on available coverage and signal strength. The server acts as a communication bridge between geographically distributed devices, enabling comprehensive information sharing across the

entire charging network. The following figure demonstrates the real-time data exchange captured on the EEBLAB ThingsBoard local server implementation:

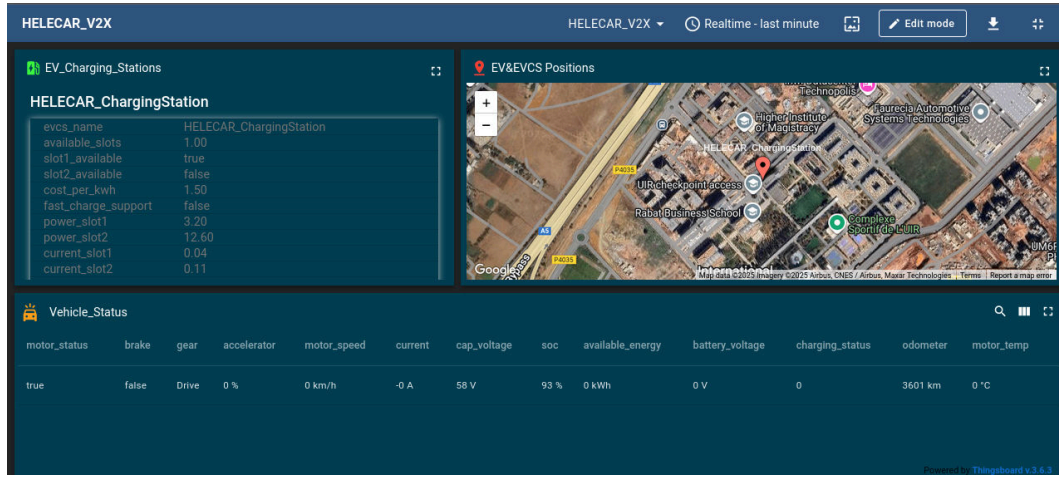


Figure 4.5: EVCS and Vehicle Data Transmitted to EEBLAB ThingsBoard Local Server

When the vehicle arrives at the charging facility and the charging connector is physically connected to the EVCS slot, short-range communication via the ESP-NOW protocol is automatically initiated between the ESP32 modules installed on both the RSU and OBU systems. This transition is triggered by current flow detection algorithms: the EVCS detects current levels exceeding 0.5A indicating vehicle connection, while the vehicle monitoring system detects charging current above 30A confirming active power transfer. Upon detection of these current thresholds, both systems automatically update their operational status to "CHARGING" mode.

The proximity-based activation system calculates the distance between the vehicle and charging station using the Haversine formula applied to GPS coordinates from both systems. Once the ESP32 modules establish wireless pairing and physical connection is confirmed, systematic session data exchange commences at five-second intervals. The Twizy OBU transmits critical vehicle parameters to the EVCS-RSU, including:

- Real-time State of Charge (SOC) updates
- Battery voltage and current measurements
- Continuous position monitoring for safety verification
- Charging termination requests based on user preferences or system conditions

Concurrently, the ESP32 module within the EVCS-RSU provides essential charging session information to the vehicle system:

- Cumulative energy delivered during the current session
- Real-time cost calculations based on energy consumption and pricing algorithms
- Session duration tracking for billing and operational purposes
- Grid status updates including power availability and system alerts

During active charging sessions, the system implements autonomous session management based on multiple operational parameters. SOC monitoring tracks progress toward target levels (configurable to 80%, 90%, or 100% as desired by the user). Cost threshold monitoring prevents unexpected charges by terminating sessions when predetermined spending limits are reached. Safety parameter checking continuously monitors electrical values to ensure operation within safe operating boundaries. Session timeout management prevents indefinite charging sessions and ensures charging station availability for other users.

The system continuously updates cost variables to simulate real-world pricing scenarios while tracking charging duration for billing purposes. Charging sessions terminate automatically when any of the following conditions are met: SOC reaches the desired level, cost exceeds the predetermined threshold (configured to 10.00 units in the current implementation), session timeout occurs, or safety fault detection triggers emergency shutdown. Manual session termination is available through transmission of a `stop_charging: TRUE` command from the vehicle, which the EVCS confirms before terminating power delivery when current drops below the detection threshold.

Upon session completion, final data exchange includes comprehensive session summary information:

- Total energy delivered measured in kilowatt-hours (kWh)
- Total session cost calculated in monetary units
- Complete session duration recorded in minutes
- Final SOC percentage achieved
- Session completion status confirmation

System cleanup procedures include comprehensive session logging to CSV files for historical analysis, automatic marking of the charging slot as available for subsequent users, and reduction of V2G communication to discovery mode to conserve system resources while maintaining readiness for new charging sessions.

State Transitions

The V2G communication system operates through a well-defined state machine that manages the progression from vehicle discovery through charging completion. Figure 4.6 illustrates the complete state transition diagram for the V2G communication protocol.

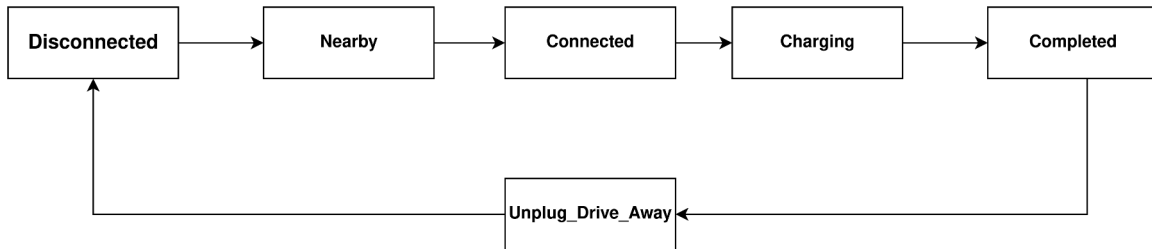


Figure 4.6: V2G Communication State Transition Diagram

The state machine progresses through six distinct operational states: **DISCONNECTED** represents the initial state when no communication exists between vehicle and charging station; **NEARBY** indicates proximity-based discovery when vehicles enter the detection range; **CONNECTED** confirms establishment of communication links; **NEGOTIATING** handles parameter exchange and session setup; **CHARGING** manages active power transfer and monitoring; and **COMPLETED** finalizes session data and prepares for system reset. The system returns to **DISCONNECTED** state when vehicles leave the charging area or disconnect from the charging infrastructure.

4.5 Experimental Results and System Validation

The implementation testing successfully demonstrated bidirectional communication between the custom EVCS and the Renault Twizy electric vehicle. The experimental validation confirmed proper operation of all system components and communication protocols under realistic operating conditions. Figure 4.7 demonstrates the ESP32 module on the Twizy operating correctly in standby mode, awaiting physical connection of the charging cable. The system simultaneously broadcasts vehicle data to surrounding infrastructure and other vehicles, enabling Vehicle-to-Vehicle (V2V) communication capabilities. The display clearly indicates the desired SOC set to 100%, vehicle readiness for charging

operations, and current charging status showing "CHARGING: NO" confirming the pre-connection state. On the other hand we can notice on that the EVCS system's charging status which is the second column coming right after EELAB_EVSC_slot_1 is equal to 0, since no load has been plugged yet to the slot figure 4.8.

```

=====
ESP-NOW Serial: /dev/ttyUSB1
CAN Data Input: can_data_snapshot.csv
ESP-NOW CSV: ESP_now_data_20250828_0939.csv
CAN+GPS CSV: merged_can_data_20250828_0939.csv
V2G Sessions: v2g_sessions_20250828_0939.csv
Features: Real CAN Battery Data + Simple Current-Based V2G Detection
Charging Detection: Current > 0.5A
=====

Simple V2G System Active!
Now sending REAL battery data from CAN:
  • Voltage from CAN 0x425 (Battery Management)
  • Current from CAN 0x155 (Battery Status)
  • Energy from CAN 0x425 (Available Energy)
  • Charging Status from CAN 0x425 (Protocol Status)
V2G will start when:
  1. Vehicle is plugged in (physical connection)
  2. Current > 0.5A detected (charging starts)
Monitoring for charging sessions...

Sending V2V/V2G CAN data to ESP32
Simple CAN Data Reader with Charging Detection
Monitoring for charging sessions based on current change...

Sent V2V/V2G data [10]: SOC=68%, V=0.0V, I=-1.2A, Speed=-0.0km/h
  Position: 33.986168,-6.725011 Alt:159.0m, Accel:0%, Brake:OFF, Gear:Neu
  Charging: NO, Ready: YES, Desired SOC: 100%
Sent V2V/V2G data [20]: SOC=68%, V=0.0V, I=-1.2A, Speed=-0.0km/h
  Position: 33.986164,-6.724997 Alt:157.7m, Accel:0%, Brake:OFF, Gear:Neu
  Charging: NO, Ready: YES, Desired SOC: 100%

[0] 0:~bash 1:esp32_vehicle 2:esp32_jetson 3:ssh_twizy* 4:ssh_evcs 5:V2X 6:esp32_evcs- "fedora" 09:41 28-Aug-25

```

Figure 4.7: Twizy V2G System Output Prior to Charging Session Initiation

```

EELAB_EVSE_slot_10 33.986107 -6.724805 234.3 0.02 2.3 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.3 0.02 2.3 10.696 50 0.45
EELAB_EVSE_slot_10 33.986107 -6.724805 234.4 0.02 2.2 10.696 50 0.45
EELAB_EVSE_slot_10 33.986107 -6.724805 234.3 0.02 2.2 10.696 50 0.45
EELAB_EVSE_slot_10 33.986107 -6.724805 234.2 0.02 2.2 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.2 0.02 2.2 10.696 50 0.45
EELAB_EVSE_slot_10 33.986107 -6.724805 234.1 0.02 2.2 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.1 0.02 2.2 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.0 0.02 2.2 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 233.9 0.02 2.4 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 233.9 0.02 2.2 10.696 50 0.41
EELAB_EVSE_slot_10 33.986107 -6.724805 234.0 0.02 2.2 10.696 50 0.41
EELAB_EVSE_slot_10 33.986107 -6.724805 234.1 0.02 2.2 10.696 50 0.41
EELAB_EVSE_slot_10 33.986107 -6.724805 234.2 0.02 2.2 10.696 50 0.41
EELAB_EVSE_slot_10 33.986107 -6.724805 234.3 0.02 2.2 10.696 50 0.41
EELAB_EVSE_slot_10 33.986107 -6.724805 234.3 0.02 2.2 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.4 0.02 2.2 10.696 50 0.41
EELAB_EVSE_slot_10 33.986107 -6.724805 234.3 0.02 2.4 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.1 0.02 2.4 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.3 0.02 2.3 10.696 50 0.41
EELAB_EVSE_slot_10 33.986107 -6.724805 234.4 0.02 2.2 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.3 0.02 2.3 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.3 0.02 2.2 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.5 0.02 2.2 10.696 50 0.41
EELAB_EVSE_slot_10 33.986107 -6.724805 234.5 0.02 2.2 10.696 50 0.43
EELAB_EVSE_slot_10 33.986107 -6.724805 234.6 0.02 2.4 10.696 50 0.44
EELAB_EVSE_slot_10 33.986107 -6.724805 234.9 0.02 2.2 10.696 50 0.43

[0] 0:~bash 1:esp32_vehicle 2:esp32_jetson 3:ssh_twizy* 4:ssh_evcs 5:V2X 6:esp32_evcs- "fedora" 09:55 28-Aug-25

```

Figure 4.8: EVCS System Output Prior to Charging Session Initiation

Figure 4.9 confirms successful charging session initiation following physical cable connection. The ESP32 modules from both the EVCS-RSU and vehicle OBU established successful data exchange, with the vehicle system successfully receiving critical EVCS parameters including MAC address, geographic coordinates, output current and voltage specifications, and operational status information. Importantly, the ESP32 module continues broadcasting to surrounding vehicles and infrastructure during the charging session, demonstrating simultaneous V2G and V2V communication capabilities.

Figure 4.10 validates successful bidirectional communication from the charging station perspective, confirming receipt of vehicle information and establishment of the complete V2G communication link. The experimental results demonstrate successful implementation of all designed V2G communication protocols and validate the system's readiness for practical deployment in smart grid applications.

```

CHARGING SESSION STARTED
Detected by current change: 32.8A
Session Start: SOC=68%, Energy=0.0kWh
Voltage: 0.0V, Current: 32.8A
=====
[CHARGING] Duration: 0m | SOC: 68% | Energy: 0.0kWh | Current: 32.8A
[V2G] EVCS Data Received - MAC: 4C:AE:B3:84:30:00
[V2G] Session: Energy=0.000kWh, Cost=0.00, Rate=0.00/kWh, ID=0, Available=NO
[V2G] EVCS Position: 33.986107, -6.724805, Voltage=225.9V, MaxCurrent=13.0A
[CHARGING] Duration: 0m | SOC: 68% | Energy: 0.0kWh | Current: 32.5A
[CHARGING] Duration: 0m | SOC: 68% | Energy: 4.1kWh | Current: 32.8A
[V2G] EVCS Data Received - MAC: 4C:AE:B3:84:30:00
[V2G] Session: Energy=0.005kWh, Cost=0.00, Rate=0.00/kWh, ID=0, Available=NO
[V2G] EVCS Position: 33.986107, -6.724805, Voltage=226.2V, MaxCurrent=13.0A
Sent V2V/V2G data [40]: SOC=68%, V=57.5V, I=32.8A, Speed=0.0km/h
Position: 33.986198, -6.724993 Alt:152.4m, Accel:0%, Brake:0FF, Gear:
Charging: YES, Ready: YES, Desired SOC: 100%
[0] 0:~bash 1:esp32_vehicle 2:esp32_jetson 3:ssh_twizy* 4:ssh_evcs 5:V2X 6:esp32_evcs- "fedora" 09:41 28-Aug-25

```

Figure 4.9: Twizy V2G System Output Following Charging Session Initiation

```

EELAB_EVSE_slot_11 33.986107 -6.724805 226.1 9.38 2102.0 10.473 50 0.99
EELAB_EVSE_slot_11 33.986107 -6.724805 226.0 9.38 2100.6 10.474 50 0.99
[V2G] EVCS data sent to vehicle
[V2G] Session Energy: 0.009 kWh | Cost: 0.01
[V2G] Data sent to vehicle successfully
[V2G] Vehicle MAC: A0:B7:65:59:56:38
[V2G] Vehicle Position: 33.986198, -6.724993 (GPS: Valid)
[V2G] Battery: SOC=68%, Voltage=57.5V, Current=32.8A
[V2G] Distance to vehicle: 20.2 meters
[V2G] Vehicle MAC: A0:B7:65:59:56:38
[V2G] Vehicle Position: 33.986198, -6.724993 (GPS: Valid)
[V2G] Battery: SOC=68%, Voltage=57.5V, Current=32.8A
[V2G] Distance to vehicle: 20.2 meters
EELAB_EVSE_slot_11 33.986107 -6.724805 226.0 9.38 2100.7 10.475 50 0.99
EELAB_EVSE_slot_11 33.986107 -6.724805 225.9 9.38 2099.7 10.477 50 0.99
[V2G] EVCS data sent to vehicle
[V2G] Session Energy: 0.012 kWh | Cost: 0.02
[V2G] Data sent to vehicle successfully
[V2G] Vehicle MAC: A0:B7:65:59:56:38
[V2G] Vehicle Position: 33.986198, -6.724993 (GPS: Valid)
[V2G] Battery: SOC=68%, Voltage=57.5V, Current=32.8A
[V2G] Distance to vehicle: 20.2 meters
[V2G] Vehicle MAC: A0:B7:65:59:56:38
[V2G] Vehicle Position: 33.986198, -6.724993 (GPS: Valid)
[V2G] Battery: SOC=68%, Voltage=57.5V, Current=32.8A
[V2G] Distance to vehicle: 20.2 meters
EELAB_EVSE_slot_11 33.986107 -6.724805 225.9 9.38 2099.5 10.478 50 0.99
[0] 0:~bash 1:esp32_vehicle 2:esp32_jetson 3:ssh_twizy* 4:ssh_evcs 5:V2X 6:esp32_evcs- "fedora" 09:41 28-Aug-25

```

Figure 4.10: EVCS System Output During Active Charging Session

```

[V2G] EVCS Data Received - MAC: 4C:AE:B3:84:30:00
[V2G] Session: Energy=0.201kWh, Cost=0.00, Rate=0.00/kWh, ID=0, Available=NO
[V2G] EVCS Position: 33.986107, -6.724805, Voltage=224.4V, MaxCurrent=13.0A
[CHARGING] Duration: 5m | SOC: 72% | Energy: 0.0kWh | Current: 32.2A
[CHARGING] Duration: 5m | SOC: 72% | Energy: 0.0kWh | Current: 32.0A
[V2G] EVCS Data Received - MAC: 4C:AE:B3:84:30:00
[V2G] Session: Energy=0.207kWh, Cost=0.00, Rate=0.00/kWh, ID=0, Available=NO
[V2G] EVCS Position: 33.986107, -6.724805, Voltage=226.0V, MaxCurrent=13.0A

CHARGING SESSION COMPLETED
Duration: 5 minutes
SOC: 68% → 72% (+4%)
Energy: 0.0 → 0.0kWh (+0.0kWh)
=====
Sent V2V/V2G data [160]: SOC=72%, V=57.5V, I=-0.8A, Speed=0.0km/h
Position: 33.986191, -6.724975 Alt:138.2m, Accel:0%, Brake:0FF, Gear:
Charging: NO, Ready: YES, Desired SOC: 100%
Sent V2V/V2G data [170]: SOC=72%, V=0.0V, I=-0.8A, Speed=0.0km/h
Position: 33.986183, -6.724958 Alt:139.6m, Accel:0%, Brake:0FF, Gear:
Charging: NO, Ready: YES, Desired SOC: 100%
Sent V2V/V2G data [180]: SOC=72%, V=0.0V, I=-0.8A, Speed=0.0km/h
Position: 33.986179, -6.724951 Alt:142.9m, Accel:0%, Brake:0FF, Gear:
Charging: NO, Ready: YES, Desired SOC: 100%
Sent V2V/V2G data [190]: SOC=72%, V=0.0V, I=-0.8A, Speed=0.0km/h
Position: 33.986198, -6.724963 Alt:141.2m, Accel:0%, Brake:0FF, Gear:
Charging: NO, Ready: YES, Desired SOC: 100%
[0] 0:~bash 1:esp32_vehicle 2:esp32_jetson 3:ssh_twizy* 4:ssh_evcs 5:V2X 6:esp32_evcs- "fedora" 09:49 28-Aug-25

```

Figure 4.11: Twizy V2G System Output When Charging Session Has ended


```

EELAB_EVSE_slot_11 33.986107 -6.724805 232.1 0.53 65.0 10.675 50 0.53
[V2G] EVCS data sent to vehicle
[V2G] Session Energy: 0.000 kWh | Cost: 0.00
[V2G] Data sent to vehicle successfully
[V2G] Vehicle MAC: A0:B7:65:59:56:38
[V2G] Vehicle Position: 33.986183, -6.724954 (GPS: Valid)
[V2G] Battery: SOC=72%, Voltage=0.0V, Current=-0.8A
[V2G] Distance to vehicle: 16.3 meters
[V2G] Vehicle MAC: A0:B7:65:59:56:38
[V2G] Vehicle Position: 33.986183, -6.724952 (GPS: Valid)
[V2G] Battery: SOC=72%, Voltage=57.5V, Current=-0.8A
[V2G] Distance to vehicle: 16.1 meters
[V2G] CHARGING SESSION COMPLETED
[V2G] Energy: 0.000 kWh, Cost: 0.00, Duration: 0 min
EELAB_EVSE_slot_10 33.986107 -6.724805 232.2 0.28 12.6 10.675 50 0.19
[V2G] CHARGING SESSION STARTED - ID: 135
EELAB_EVSE_slot_11 33.986107 -6.724805 226.8 8.64 1900.0 10.675 50 0.97
[V2G] EVCS data sent to vehicle
[V2G] Session Energy: 0.000 kWh | Cost: 0.00
[V2G] Data sent to vehicle successfully
[V2G] Vehicle MAC: A0:B7:65:59:56:38
[V2G] Vehicle Position: 33.986183, -6.724952 (GPS: Valid)
[V2G] Battery: SOC=72%, Voltage=57.5V, Current=-0.8A
[V2G] Distance to vehicle: 16.1 meters
EELAB_EVSE_slot_11 33.986107 -6.724805 226.2 9.71 2176.4 10.677 50 0.99
[V2G] Vehicle MAC: A0:B7:65:59:56:38
[V2G] Vehicle Position: 33.986172, -6.724949 (GPS: Valid)
[V2G] Battery: SOC=72%, Voltage=57.5V, Current=-0.8A
[V2G] Distance to vehicle: 15.1 meters

```

Figure 4.12: EVCS System Output After Active Charging Session Has ended

The experimental validation achieved a communication range up to proximately 107 meters for ESP-NOW protocol, with message latency averaging 23 milliseconds during active sessions, but the successful received by both systems drops significantly after supassing about 40 meters, and the latency starts increasing averaging 1.5s to 2s. The distance calculation accuracy using GPS coordinates demonstrated precision within 0.5 meters when compared to physical measurements, validating the proximity-based activation system.

It is also worth mentioning that we have already created other charging points for the vehicle and installed other EVCS_RSU (which work the same way as the one on the charging station of the EEBLAB grid and can also send informations relative to its slots availabilities, locations, charging power, ...etc), and the reason for that is meanly to simulate the EVCS_Recommendation_System figure 4.13 and Speed Forecasting when they are ready to be implemented on the vehicle.

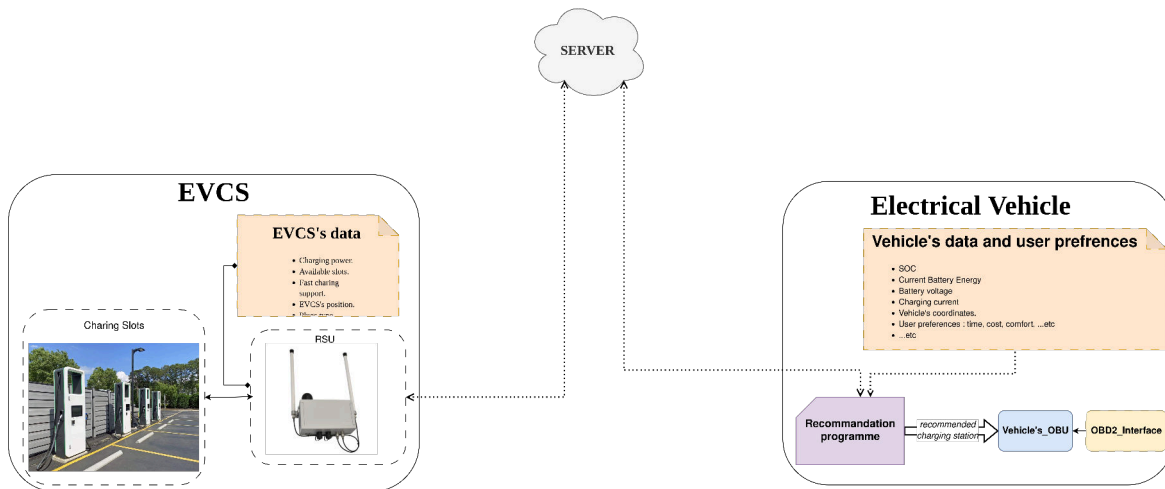


Figure 4.13: V2G Recommendation System Architecture

4.5.1 Challenges and limitations

Despite the successful implementation of the communication framework between the Twizy and the custom charging station, there were technical and practical constraints that limited the full realization of the V2G concept:

- **Bidirectional Power Flow:** Although our architecture was designed with bidirectional interaction in mind, we were not able to implement actual power feedback from the Twizy battery

to the grid. This was mainly due to hardware constraints (e.g., lack of a certified bidirectional charger/inverter) and safety considerations. As a result, the current implementation focused on data-level bidirectionality (information exchange) rather than energy-level bidirectionality.

- **Standard Compliance:** Our system relied on ESP-NOW and custom communication protocols rather than fully adopting standards such as ISO 15118. While this approach was sufficient for prototyping, it limits interoperability with commercial EVs and charging stations. The lack of ISO 15118 compliance was primarily due to the complexity of the standard, making a full implementation extremely challenging within the timeframe and scope of an internship project.
- **Lack of necessary equipments:** Developing a fully functional V2G system requires the integration of multiple vehicles and charging stations to evaluate large-scale data exchange, synchronization, and coordinated charging. Due to limited resources, we were restricted to a single-vehicle and single-station setup.
- **Testing Environment Limitations:** All tests were conducted within the University premises (UIR), which constrained the evaluation of the system's performance in larger, real-world geographical deployments. As a result, aspects such as long-range communication reliability and interoperability with public charging infrastructure could not be fully assessed.

4.6 Chapter Summary

In this chapter, we presented the implementation of a Vehicle-to-Grid (V2G) communication system using the Renault Twizy as a test platform. The prototype successfully demonstrated bidirectional information exchange and integration with a custom charging station. However, actual bidirectional power transfer was not achieved due to hardware and safety constraints. These results highlight both the feasibility and the challenges of V2G systems, and serve as a foundation for future improvements. Future work could explore the following directions:

- Integration of certified bidirectional chargers to enable true energy feedback.
- Implementation of ISO 15118 to ensure interoperability with commercial EVs and charging stations.
- Testing with multiple vehicles and charging stations to assess scalability and synchronization.
- Real-world deployment beyond the laboratory environment.
- Development of an advanced charging station recommendation system based on driver preferences.

Conclusion and Future Work

General Conclusion

This project aimed to design, implement, and validate a custom **V2X** system on a Renault Twizy electric vehicle, providing a low-cost and scalable research platform for connected mobility applications. Throughout this internship, significant progress was made in both hardware and software development, resulting in a functional proof-of-concept system. The contributions of this work include the development of a custom **On-Board Unit (OBU)** based on the Jetson Nano and a CAN interface for vehicle data acquisition, as well as the implementation of **(V2V)** communication using ESP-NOW to enable the broadcast and reception of basic safety messages between multiple nodes. In addition, a **(V2N)** connection was established, allowing real-time transmission of vehicle status data to a local ThingsBoard server, where a dedicated dashboard was created to visualize key parameters such as speed, battery state of charge, and operational status. Finally, a preliminary **(V2G)** proof-of-concept was designed, demonstrating the exchange of information between the Twizy and a custom-built Electric Vehicle Charging Station (EVCS). Overall, this work demonstrates the feasibility of developing a research-oriented V2X platform with accessible hardware and open-source tools, laying the groundwork for future standard-compliant implementations.

Limitations & Challenges

Despite the successful outcomes of this project, several limitations and challenges were encountered that impacted the overall implementation. **Hardware constraints** played a significant role, as the use of prototyping boards limited processing power, wireless communication range, and data logging frequency, affecting real-time performance. In terms of **(V2G) implementation**, true bidirectional power flow could not be achieved due to the absence of a certified bidirectional charger; the system focused solely on **data exchange and control signaling**. Additionally, the project faced **scalability challenges**, as testing was limited to a single Twizy and a small number of ESP32 nodes, which may not fully represent performance in large-scale vehicular networks. Another critical limitation involved **protocol compliance**, since the current implementation relies on ESP-NOW and custom messaging, which are not fully aligned with industrial V2X communication standards such as DSRC or C-V2X. Finally, **environmental factors**, including real-world wireless interference and occasional packet loss, affected communication reliability, highlighting the need for robust error-handling mechanisms to ensure consistent performance.

Future Work

To advance this project and align it with industrial and academic applications, several directions are recommended. **Standardization and compliance** should be prioritized by integrating recognized communication stacks such as DSRC or C-V2X for V2V/V2I communication, and implementing ISO 15118 to enable secure plug-and-charge operations in V2G scenarios. From a **hardware perspective**, upgrading the OBU is essential to achieve real-time data processing and reduce communication

latency, and integrating a certified bidirectional charger would enable true vehicle-to-grid energy transfer. **Scalability testing** should be conducted by deploying the system across multiple vehicles and roadside units, allowing analysis of network performance under high load and realistic mobility patterns. Additionally, **enhanced V2N features** can be developed by expanding the ThingsBoard dashboard to incorporate predictive analytics, energy management, and preventive maintenance alerting mechanisms. Finally, exploring **AI and data-driven applications** such as intelligent energy optimization algorithms and potential healthcare-related use cases (e.g., driver fatigue detection) could significantly increase the system's value and impact. Overall, addressing the current limitations and implementing these improvements will enable the evolution of the system into a fully standardized, robust, and scalable V2X platform suitable for both industrial and research environments.

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